IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 12 November 2024, accepted 13 December 2024, date of publication 30 December 2024, date of current version 22 January 2025.

Digital Object Identifier 10.1109/ACCESS.2024.3524261

SURVEY

A Review of Causal Methods for High-Dimensional Data

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This work was supported by the University of Oulu and the Academy of Finland Profi under Grant 326291.

ABSTRACT Causal learning from observational data is an important scientific endeavor, but the statistical and computational challenges posed by the high-dimensionality of many modern datasets are substantial. Peculiarities such as spurious correlations, endogeneity, noise accumulation, and deflated empirical covariance estimation complicate analysis. These issues may lead to confounding bias, which can be misleading when attempting to learn the true causal relationships and causal effects between variables. In this survey, we provide a comprehensive review of causal analysis and the theory behind high-dimensionality. We discuss the effects of high-dimensionality on causal estimation methods and their corresponding solutions. Finally, we present evaluation metrics and software tools for both causal effect estimation and causal discovery.

INDEX TERMS Causal discovery, causal effect estimation, causal methods, confounding bias, endogeneity, high-dimensionality.

I. INTRODUCTION

In our daily life, we assess both positive (credit) and negative (regret) outcomes related to our experiences. By analysing these evaluations, we learn which actions lead to favorable decisions. This intuitive process of evaluating and learning from our experiences is a form of decision theory that is deeply connected to the concept of causality.

Causality is a fundamental idea in scientific inquiry that seeks to understand the relationship between variables, or whether and to what extent changes in one variable cause changes in another variable. The field has gained significant attention in recent years due to its importance in various domains, including statistics, epidemiology, social sciences, and computer science [1], [2], [3], [4], [5], [6].

Causal learning can be broadly categorized into two branches: causal inference and causal discovery. Causal inference focuses on estimating the causal effect of an intervention or treatment on an outcome. The goal is to quantify the effect of changing one variable on another. However, causal discovery focuses on identifying causal structures or patterns in an observational dataset. Its goal is to build a causal model, often represented as a directed acyclic graph (DAG) that reflects the underlying causal structure or relationship between variables. There is a clear relationship between them: causal discovery provides insights into the causal structure, which can inform and guide causal inference.

Causal learning is typically achieved through well-designed randomized experiments. However, since randomized experiments are not always feasible for ethical or practical reasons, observational studies are an alternative. Athey et al. [7] and Ghassami et al. [8] proposed a method to identify and estimate the causal effect of a treatment on a long-term outcome using data from both observational and experimental domains when available. Here, we focus on observational studies. In causal analysis, we typically investigate the effects of a given treatment or predictor (X) on the outcome of interest (Y) in the presence of covariates (W). As a motivational example, consider that a researcher has an observational dataset containing the variables: {Qualification, Hiring, Bribe}. The researcher designs the study, deciding on the effect of one variable on another to investigate, as shown in Fig. 1. This figure

The associate editor coordinating the review of this manuscript and approving it for publication was Diego Bellan.

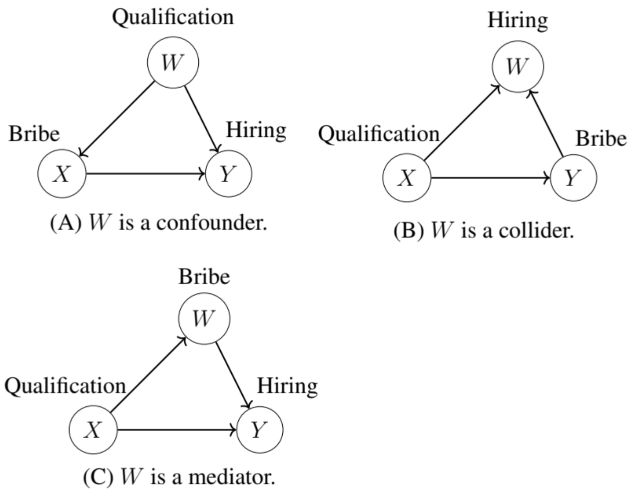

FIGURE 1. Basic DAG that shows the relationship between treatment X, outcome Y, and covariate W.

illustrates the relationships of covariate W with the treatment X and the outcome Y. Covariate W can be a confounder if it influences both X and Y (Fig. 1A), a collider if it is influenced by both X and Y (Fig. 1B), or a mediator if it is influenced by X and also influences Y (Fig. 1C).

In the scenario depicted in Fig. 1A, a researcher aims to study the direct causal effect of the variable Bribe on the variable Hiring. Here, the variable Qualification directly influence Hiring; however, candidates with lower qualifications may feel compelled to offer a bribe to increase their chances of being hired. Hence, there is an association represented as X : Bribe $\leftarrow$ W : Qualification $\rightarrow$ Y : Hiring, which introduces a spurious correlation. The phrase "Correlation does not imply causation," which we often hear, arises because of such spurious correlations. This means that the researcher needs to adjust for the association or confounding effect of the qualification in this particular context. This idea is humorously illustrated in an xkcd comic,[1] where characters engage in a dialogue about their misconceptions regarding correlation and causation. This exchange highlights the critical distinction between the two concepts: while correlation may suggest a relationship between variables, it does not confirm a direct causal link.

In Fig. 1B, suppose that a researcher wants to estimate the direct causal effect of qualification on bribe. Here, both qualification and bribe directly cause hiring. In this case, one can estimate the direct causal effects of Qualification on Bribe without adjustment since there is no backdoor path that introduces a spurious correlation from Qualification to Bribe.

In Fig. 1C, a researcher wants to study the direct causal effects of Qualification on Hiring. However, there is a causal path from $X \rightarrow W \rightarrow Y$ that requires further investigation. The DAG and its related concepts are thoroughly discussed in subsection (II-A).

[1]https://www.xkcd.com/552/

In causal studies, variable selection or adjustment for covariates is a challenging task because traditional statistical measures, such as p-values or correlations with other variables, are often insufficient. For specific examples, please refer to [9]. Consequently, several common estimation biases can arise during covariate selection. These include confounding bias, which occurs when a confounder is either unaccounted for or omitted from the analysis; mediator bias, which happens when a mediator is omitted even though the direct effect is of interest; and collider bias, which arises when a collider is mistakenly included in the analysis. It is important to note that no single paper can encompass all aspects of causality. This survey specifically focuses on confounding bias, which represents a significant obstacle to estimating causal effects and uncovering causal relationships in observational studies.

Based on the types of confounders, different methods and algorithms have been developed to estimate causal effects and discover causal relationships. When all confounders are observable, several methods can be used to estimate the causal effect of treatment on the outcome, including outcome regression with the g-formula [10], inverse probability weighting (IPW) from propensity scores [11], and double robust methods such as augmented IPW [12], [13] and targeted maximum likelihood estimation (TMLE) [14].

Austin [11] offers a detailed review of propensity score methods, including their strengths and limitations. The instrumental variables (IVs) method is a powerful tool for causal inference in the presence of unobserved confounders [5], [15], [16], but careful attention to the assumptions and validity of the IVs is required.

Causal inference in high-dimensional settings involves a large number of covariate variables, along with many potential confounders, whether observed, unobserved, or both. Direct control of all covariates becomes infeasible; hence, it relies on high-dimensional statistical inference methods, such as dimension reduction, regularization, or effective variable selection techniques. For statistical inference in high-dimensional settings, we recommend that interested readers refer to [17], [18], [19], [20], [21], and [22].

However, there are some potential problems with regularization methods. For example, in cases where variables are highly correlated, the LASSO regularization method may randomly select variables [23]. When two or more variables are highly correlated due to high-dimensionality, LASSO may choose one variable over another. More specifically, when confounders are strongly correlated with the treatment but less correlated with the outcome, the LASSO method may ignore selecting the confounder, leading to substantial biases. Hence, de-biasing techniques in high-dimensional inference are essential; see for example the recent developments in [24], [25], [26], and [27].

In high-dimensional observational data, one convenient way to estimate the causal treatment effect between the treated and control groups is to investigate the effects of a treatment on the outcome at the population level. The main task here is to balance the covariate distribution between the two groups. This means that, in addition to regularization methods, achieving covariate balance is essential. For further details, see [28], [29], [30], and [31].

Causal inference has plausible applications in decision-making and counterfactual reasoning, enabling the assessment of outcomes that would have occurred under alternative scenarios. For example, scientists use genome-wide association studies (GWAS) to understand how genes determine phenotypic traits [32]. Doctors aim to understand how medical treatments affect disease progression [33]. Policymakers seek to learn how social programs impact social outcomes [34]. Recommender systems analyse users' rating data to identify preferences, predict how users would rate items they have not interacted with, and suggest items based on those predictions [35], [36]. Marketers aim to measure whether advertising campaigns increase clicks or boost sales [37], while educators compare different teaching methods to determine which improves student outcomes [38].

A large number of books and review papers have been published on causal learning. Among these, Pearl [4] and Pearl et al. [6] comprehensively cover the theory of causality that integrates probabilistic, manipulative, counterfactual and structural approaches. Imbens and Rubin [5] offer an extensive approach to causal inference, addressing both experimental and observational data. They outline the necessary assumptions and discuss prominent analysis methods, such as propensity score methods and instrumental variables. Morgan and Winship [39] introduce the concepts of counterfactual theory and causal inference, including causal graphs. The book by Peters et al. [40] provides detailed coverage of topics such as causal discovery, structural learning, and structural causal models. The book by Hernán and Robins [10] explores causal inference, emphasizing precise question formulation and including practical examples aimed at researchers across various fields.

The review paper by Pearl [3] emphasizes important advancements in causal inference and advocates for a transition from traditional statistical analysis to causal analysis of multidimensional data, whereas the survey paper by Guo et al. [41] is a review of methods for causal learning with connections to machine learning. Yao et al. [42] offer a survey of causal inference methods within the potential outcome framework. Schölkopf et al. [43] present the concepts of causal inference and their relation to machine learning, highlighting the potential of causality to contribute to modern machine learning research. The survey by Cheng et al. [44] provides a comprehensive review of evaluation protocols for causal learning algorithms.

The approaches for learning causal structure, also called causal discovery from data, are discussed in the review papers by Heinze-Deml et al. [45], Vowels et al. [46], and Squires and Uhler [47]. These approaches include constraint-based methods [1], score-based methods [48] and functional causal models [49]. Nogueira et al. [50] review methodologies for both causal discovery and inference.

The review papers on causality mentioned above hesitate to consider the effect of high-dimensionality on causal estimation methods. As a result, our survey makes a threefold contribution: First, we provide a comprehensive review of causal analysis and the theory behind high-dimensionality. Second, we discuss the effects of high-dimensionality on causal estimation methods and their corresponding solutions. Finally, we present evaluation metrics and software packages for both causal effect estimation and causal discovery focusing specifically on high-dimensional data.

II. CAUSAL INFERENCE FRAMEWORKS AND METHODS

The two most common types of causal inference frameworks are structural causal models (SCMs) and potential outcomes (PO) models. In subsequent sections, we review both frameworks and their associated causal estimation methods.

In our notation, we use uppercase letters to denote random variables, lowercase letters to denote particular values of corresponding random variables, bold lowercase letters to denote vectors and bold uppercase letters to denote matrices.

A. STRUCTURAL CAUSAL MODELS FRAMEWORK

Structural causal models (SCMs) describe relevant features and their interactions in a system. An SCM is composed of a set of variables and a set of functions that determine the values of these variables [4], [6]. Formally, a probabilistic SCM $\mathcal{M}$ consists of a set of independent random variables U with a probability function $P(u)$ defined over the domain of U, a set of functions $F = \{f_1, \ldots, f_p\}$, and a set of variables $X = \{X_1, \ldots, X_p\}$ such that $X_j = f_j(\mathrm{PA}_j, U_j),\ \forall j = 1, \ldots, p$, where p represents the number of variables in the model. Here, $\mathrm{PA}_j \subseteq X \setminus \{X_j\}$ is the subset of X that consists of the parents of X_j. This can be represented as a four-tuple: $\mathcal{M} = \langle U, X, F, P(u)\rangle$. Hence, the probablity function $P(u)$ and the functions F determine the distribution of $\mathcal{M}$. A submodel $\mathcal{M}_X = \langle U, X, F_x, P(u)\rangle$ is a modified version of the original model $\mathcal{M}$ that accounts for an intervention on X, where the set F_x encapsulates the relevant causal relationships in the model while explicitly conditioning on the values of the observed variables $X_1, X_2, \ldots, X_p$. This concept is crucial for understanding how interventions alter the relationships in the system.

The SCM concept is associated with a graphical causal model where nodes denote variables, while edges between the nodes represent relationships between these variables. This causal concept is illustrated using a graphical causal model known as a DAG. Fig. 2 shows a basic DAG illustrating the relationship between the observed variables X, Y and W, along with the unobserved variable U. For completeness, one can add the background variables U_X, U_Y, and U_W to Fig. 2, assuming that they are independent of each other. Specifically, these background variables are also called noise terms [40]. Assuming that the variables X, W, and Y are linearly related to their ancestors, the structural equations

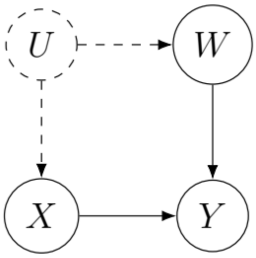

FIGURE 2. DAG representing the relationship between observed (measured) variables X, W, Y, and unobserved (unmeasured) variable U. The dashed circle indicates a hidden variable or unobserved, and a dashed arrow indicates a hidden edge or path.

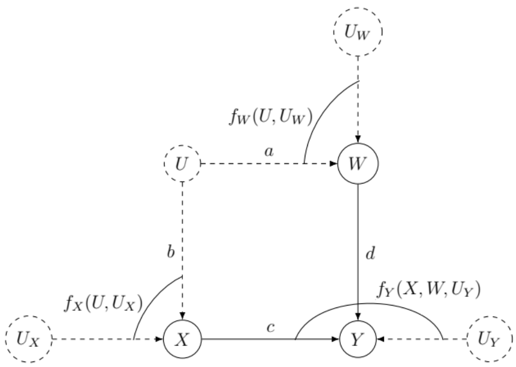

FIGURE 3. Graphical model of SCMs (1).

compatible with Fig. 3 can be written as

$$\begin{aligned} X &:= f_X(U, U_X) = bU + U_X, \\ W &:= f_W(U, U_W) = aU + U_W, \\ Y &:= f_Y(X, W, U_Y) = cX + dW + U_Y, \end{aligned} \tag{1}$$

where a, b, c, and d are path coefficients that represent the direct causal effects of the corresponding variables.

Markovian Parents: Given a DAG G, let $X = \{X_1, \ldots, X_p\}$ be an ordered set of variables, and let $P(\boldsymbol{x})$ be the joint probability distribution over these variables. (1) A set of variables PA_j is said to be the **Markovian parents** of X_j if PA_j is the minimal set of predecessors of X_j that makes X_j conditionally independent of all its other predecessors. Formally, $\mathrm{PA}_j \subseteq \{X_1, \ldots, X_{j-1}\}$ satisfies

$$P(x_j \mid \mathrm{pa}_j) = P(x_j \mid x_1, \ldots, x_{j-1}), \tag{2}$$

and no proper subset of PA_j satisfies this condition [51]. Here, x_j and pa_j denote particular values of the corresponding variables X_j and PA_j. (2) The **Markov factorization property** with respect to the DAG G can be written as

$$P(x_1, \ldots, x_p) = \prod_{j=1}^{p} P(x_j \mid \mathrm{pa}_j), \tag{3}$$

where PA_j denotes the set of Markovian parents of X_j in the graph G. When a distribution is Markov relative to a graph, the graph encodes specific independencies within the distribution, which can be leveraged for more efficient computation or for uncovering the underlying structure of the data. Hence, every distribution satisfying (2) must decompose into the product decomposition (3).

In most graphical causal models, there will be several paths connecting two variables. The question arises whether it is possible to find a criterion or technique that can be used to estimate shared dependencies in all dataset generated by a complex graphical causal model. Interestingly, there exists a method known as "**d-separation**" that performs this task. A set of nodes S blocks (d-separate) a path $\boldsymbol{p}$ when conditioning on B if and only if (1) $\boldsymbol{p}$ has a chain of nodes $A \rightarrow B \rightarrow C$ (where B is a mediator) or a fork $A \leftarrow B \rightarrow C$ (where B is a common cause), or (2) $\boldsymbol{p}$ contains an inverted fork $A \rightarrow B \leftarrow C$ in which the collider B is not in S and no descendant of B is in S [4], [6], [52], [53].

B. CAUSAL QUANTITY AND IDENTIFICATION

One of the key aims of causal inference is to estimate the effects of an action or intervention. The action may be a policy reform, medical treatment, implementation of a social program, or economic policy [6], [54]. To study whether X causes Y, we need to answer two causal questions: (1) If we do X, what would be Y? (2) How can we make Y happen? The first question focuses on determining the causal effect of X on Y while the second focuses on identifying interventions or actions to make Y happen. An action of this kind is denoted by the mathematical operator do$(\cdot)$, and the corresponding causal quantity or estimand is $p(y|\mathrm{do}(x))$. Its effect on $\mathcal{M}$ is given by the submodel $\mathcal{M}_x$, where all factors corresponding to the variables in the intervention set X are eliminated. This can be written as

$$P(x_1, x_2, \ldots, x_p \mid \mathrm{do}(x)) = \prod_{j=1}^{p} P(x_j \mid pa_j)\ \forall j \text{ with } X_j \notin X. \tag{4}$$

This is known as the truncated product formula or g-formula [6].

Identification refers to the process of transforming causal estimands into statistical estimands, specifically from $P(y|\mathrm{do}(x))$ to $P(y|x)$. For example, is the causal quantity $P(y|\mathrm{do}(x))$ identifiable from Fig. 2? If $P(y|\mathrm{do}(x))$ is identifiable, it can be iteratively computed from the joint distribution (4) as $P(x, y, w, u) = P(u)P(w|u)P(x|u)P(y|x, w)$. By intervening on the variable X, we obtain $P(x|u) = 1$. Thus, we can express this according to $P(y, w, u|\mathrm{do}(x)) = \sum_w P(y|x, w) \sum_u P(w|u)P(u)$. Next, by marginalizing over W, we get $P(w) = \sum_u P(w|u)p(u)$. Therefore, we conclude that $P(y|\mathrm{do}(x))$ is identifiable and can be written as $P(y|\mathrm{do}(x)) = \sum_w P(w)P(y|x, w)$.

To estimate causal effects in the SCMs framework, one can apply identification methods such as the *back-door criterion*, *front-door adjustment*, *ID algorithm*, or *do-calculus*.

Back-Door Criterion: A set of variables S in a DAG G must meet two conditions to satisfy the back-door criterion relative to an ordered pair of variables (X, Y) in G. First, no node in S can be a descendant of X. Second, S must block every path between X and Y that contains an arrow into X. This can be expressed as $p(y|do(x)) = \sum_{s\in S} P(y|x,s)p(s)$ [6]. For example, in Fig. 2 the set $S = \{U, W\}$ contains a back-door path $\{X \dashleftarrow U \dashrightarrow W \rightarrow Y\}$ from X to Y. U is unobserved and one cannot condition on the unobserved variable alone. In the given path, W satisfies the back-door criteria, and the causal effect of X on Y is given as $P(y|\text{do}(x)) = \sum_{w\in S_W} P(y|x,w)P(w)$. The logic behind this example is that we block the confounding biases between X and Y via W and no new confounding bias is created.

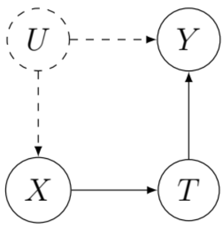

FIGURE 4. The basic example of a DAG where the back-door criterion is not applicable.

The back-door criterion offers a simple method for identifying sets of confounders that should be adjusted for. However, the criterion is not always applicable to the estimation of such effects. As an example, consider the graph model in Fig. 4. Here, the causal effect of X on Y is confounded by variable U and mediated by variable T. There is no valid back-door adjustment available for this particular case. Instead, one can use the **front-door criterion** [6]. This process involves two steps: First, obtain the causal effect of X on the mediator T. Second, obtain the causal effect of the mediator T on the outcome Y. Finally, combine both causal effects. Here, $P(t|\text{do}(x)) = P(t|x)$ does not require such adjustment since there is no backdoor path. To identify the causal effect of the mediator T on the outcome Y using the backdoor criterion, we need to condition on the variable X. We obtain $P(y|\text{do}(t)) = \sum_x P(y|t,x)P(x)$.

Combine the two previous steps to identify the causal effect of X on the outcome Y: $P(y|\text{do}(x)) = \sum_t P(t|x) \sum_{x'} P(y|t,x')P(x')$, where x stands for the values intervening on and x' stands for the values conditioning on.

In a broader sense, the identifiability of any query of the form $P(y|\text{do}(x))$ can be determined systematically employing a symbolic causal inference engine known as **do-calculus** [55], [56]. Shpitser and Pearl [57] developed the **ID algorithm** to identify conditional causal effects. The ID algorithm offers a complete algorithm and a corresponding graphical condition for the identification of all effects of the form $P(y|\text{do}(x))$ [58].

In the SCMs framework, once the post-intervention distribution $P(y|\text{do}(x))$ is known, one can estimate the average treatment effect as

$$\tau = \mathbb{E}\left(Y \mid \text{do}(x_1)\right) - \mathbb{E}\left(Y \mid \text{do}(x_0)\right), \tag{5}$$

where x_1 and x_0 are two distinct realizations of X denoting treated and control groups respectively.

C. THE POTENTIAL OUTCOMES FRAMEWORK

The potential outcome (PO) framework describes counterfactual outcomes using mathematical notation that elucidates the causal effect of a treatment on an outcome. Here, the counterfactual question is: what would have happened to outcome Y if variable X had been different from what is actually observed? It aims to explore the causal effect of X on Y considering counterfactual scenarios or alternative treatments. The PO was first proposed by Neyman [59] in his Ph.D. thesis, but only in the context of completely randomized experiments. Rubin extended it into a general framework for considering causation in both observational and experimental research, which is often referred to as the ''Rubin causal model (RCM)'' or the ''Neyman–Rubin causal model'' [60], [61], [62].

For a binary treatment $x \in \{0, 1\}$, a unit i is characterized by a pair of scalar potential outcomes denoted by $Y_i(0)$ and $Y_i(1)$. Specifically, $Y_i(0)$ represents the potential outcome for unit i if it were subjected to a control treatment $(X = 0)$, and $Y_i(1)$ represents the potential outcome for unit i if it were subjected to an active treatment $(X = 1)$ [5]. Note that $Y_i(0)$ is a counterfactual outcome if the unit actually received active treatment $(X = 1)$, and similarly, $Y_i(1)$ is a counterfactual outcome if the unit actually received the control treatment $(X = 0)$. In this context, a unit is an object that receives treatment at a particular time, whereas a treatment is an action that applies to a unit. Now we discuss the essential assumptions for establishing the credibility of causal claims in observational studies. The key assumptions are as follows.

1) THE CONSISTENCY ASSUMPTION

This states that each individual has an outcome corresponding to the observed (or factual) treatment, which can be expressed as $Y_i = XY_i(1) + (1 - X)Y_i(0)$. The individual treatment effect (ITE) can be defined as $ITE_i = Y_i(1) - Y_i(0)$. The fundamental problem of causal inference is that it is generally impossible to observe both $Y_i(1)$ and $Y_i(0)$ for the same individual simultaneously, which creates a problem akin to missing data, where the potential outcome under the control condition is unobserved [63], [64]. This issue limits direct causal inference about individual treatment effects. However, in addition to consistency, the effects of treatment can still be estimated at the population level with the following assumptions.

2) THE STABLE UNIT TREATMENT VALUE ASSUMPTION (SUTVA)

This requires that the potential outcomes of one unit are not affected by the assignment of treatment by another unit [5].

3) UNCONFOUNDEDNESS ASSUMPTION

This assumes that $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i|\boldsymbol{w}_i$, meaning that, conditional on pretreatment covariates $\boldsymbol{w}_i$, treatment assignment is comparable to random. This can be challenging in practice when adjusting for many covariates. However, using the propensity score makes this assumption more manageable, specifically $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i | \mathrm{PS}(\boldsymbol{w}_i)$ [65]. Unconfoundedness assumption is often referred to as conditional ignorability or conditional exchangeability. This assumption is typically untestable in practice, which makes sensitivity analysis essential [66], [67].

4) THE OVERLAP ASSUMPTION

Also known as the positivity assumption, this states that each individual has a positive probability of receiving each treatment level, or formally, $0 < \mathrm{PS}(\boldsymbol{w}) < 1$. This ensures sufficient overlap in the covariate distributions between the treatment groups, allowing fair comparisons between them [68].

5) THE SPARSITY ASSUMPTION

This assumes that only a small subset of parameters significantly affects the outcome, while the others are zero. This assumption is particularly useful in high-dimensional settings for controlling high-dimensional confounders, often through regularized regression adjustment [28].

Several causal treatment effects can be distinguished based on how the treatment is defined and what population is considered. By using the above assumptions, the following formulas define the parameters of the average treatment effect (ATE) [11], [69] as

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)], \tag{6}$$

and the average treatment effect on the treated (ATT) as

$$\tau^t = \mathbb{E}\left[Y_i(1) - Y_i(0) \mid X = 1\right]. \tag{7}$$

The ATE is the population-level average effect of moving an entire population from untreated to treated whereas the ATT is the average effect of treatment on those units who eventually receive treatment. The idea behind ATT is to estimate what the outcomes would have been for those who received treatment if they had not received it. It is considered easier to estimate because it requires weaker assumptions and focuses only on the treated group and their counterfactual outcomes.

To estimate ATT, we assume that for treated individuals, there exist untreated individuals with similar characteristics (covariates). This "weak overlap" condition means we only need enough untreated individuals who are comparable to the treated ones. In contrast, identifying ATE requires knowledge about both treated and untreated outcomes across the entire population. This necessitates a stronger assumption of common support (i.e. for every covariate combination, there should be a nonzero probability of both receiving and not receiving treatment). The two measures of treatment effects in randomized controlled trials coincide. This is because the treated population does not differ systematically from the overall population due to randomization. Furthermore, one can estimate the conditional average treatment effect (CATE) which describes the treatment effect at the subgroup level as

$$\tau(\boldsymbol{w}) = \mathbb{E}\left[Y_i(1) - Y_i(0) \mid \boldsymbol{w}_i = \boldsymbol{w}\right]. \tag{8}$$

CATE is a powerful concept that helps in understanding how treatment effects vary across different subgroups of a population. By focusing on specific covariates or characteristics, CATE allows for more personalized and targeted approaches in research, policy-making, and practical applications. This makes it a crucial tool for examining heterogeneous effects and tailoring interventions to specific groups.

Randomized experiments allow estimation of causal effects at the population level due to unconfoundedness [60]. When treatment X is randomly assigned based on covariates $\boldsymbol{w}$, estimators such as ATE, ATT, and CATE can be obtained using regression or propensity score methods. However, in observational studies, confounding can bias the treatment-outcome relationship due to dependence between treatment assignment and potential outcomes. In practice, it should be noted that some of the features or covariates in the p-dimensional vector $\boldsymbol{w}$ may not be common causes of both treatment X and outcome Y. An unbiased estimate of treatment effects in an observational study can be identified under strong ignorability of treatment assignment, i.e. evaluate if the unconfoundedness and overlap assumptions hold [65].

Linking SCMs and PO Frameworks: The connection between the SCMs and PO frameworks can be established using the counterfactual statement "Y would be equal to y if X had been x in unit i", written as $Y_x(i) = y$. The counterfactual $Y_x(i)$ can be written in the form of an SCM as $Y_x(i) = Y_{M_x}(i)$ (i.e., the counterfactual $Y_x(i)$ in a model $\mathcal{M}$ is the solution for Y in the submodel $\mathcal{M}_x$) [4], [70]. Furthermore, the conversion DAGs into Single-World Intervention Graphs (SWIGs), formalizing the relationship between causal DAGs and potential outcomes, is achieved through the process known as node-splitting [71]. The two frameworks rely on conditional independencies for causal identification. Consequently, they exhibit logical equivalence, meaning that solving a problem within one framework yields the same solution if approached using the other [6]. However, there is a difference between them in how problems are defined. SCMs framework uses causal graphs and mathematical equations to model the relationships between variables and their dependencies. The effects of an intervention can be estimated by controlling for confounding variables with and without the intervention. In contrast, the PO framework defines problems algebraically, using counterfactual independence assumption (ignorability) to compare individual outcomes with and without intervention, typically requiring a comparison group for causal effect estimation.

D. CAUSAL EFFECT ESTIMATION METHODS WITHOUT UNOBSERVED CONFOUNDERS

In the following subsections, we focus on observational data where X_i is a binary treatment indicator such that $X_i = 1$ if the treatment is given to the ith individual and $X_i = 0$ otherwise. The corresponding potential outcomes are $Y_i(1)$ and $Y_i(0)$, where $i = 1, 2, \ldots, n$ represents the number of individuals. Additionally, $\boldsymbol{w}_i$ is a vector of observable p-dimensional confounders. We present methods for estimating causal effects under the assumption that all confounders are observable (measurable) except for the noise terms, a condition known as the causal sufficiency assumption [72].

1) G-COMPUTATION

G-computation is a method that estimates the causal effect of a treatment on an outcome [10], [73]. The method accounts for confounding variables that may affect the relationship between exposure and outcome.

First, specify and fit an outcome regression model $\mu_{x_i}(\boldsymbol{w}_i) = \mathbb{E}[Y_i \mid X_i = x_i, \boldsymbol{w}_i]$. Then, compute the estimated values $\widehat{\mu}_1(\boldsymbol{w}_i)$ and $\widehat{\mu}_0(\boldsymbol{w}_i)$, where $\mu_1(\boldsymbol{w}_i) = \mathbb{E}[Y_i \mid X_i = 1, \boldsymbol{w}_i]$ and $\mu_0(\boldsymbol{w}_i) = \mathbb{E}[Y_i \mid X_i = 0, \boldsymbol{w}_i]$. Finally, apply the G-computation estimator to estimate the average treatment effect as

$$\widehat{\tau}_{\mathrm{G}} = \frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\mu}_1(\boldsymbol{w}_i) - \widehat{\mu}_0(\boldsymbol{w}_i)\right). \tag{9}$$

The G-computation method, also known as standardization or regression adjustment, relies on the parametric assumptions of the outcome regression model. Its accuracy estimate depends on the validity of these assumptions and often requires bootstrap methods to estimate standard errors and confidence intervals [73]. This can be computationally intensive and time-consuming, especially when dealing with complex models and large datasets.

2) INVERSE PROBABILITY OF TREATMENT WEIGHTING (IPTW)

IPTW produces a synthetic sample with covariates [11]. IPTW is an easier algorithm that utilizes unconfoundedness [74], [75], [76]. ATE can be estimated using IPTW as

$$\widehat{\tau}_{\mathrm{IPTW}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i Y_i}{\widehat{\mathrm{PS}}(\boldsymbol{w}_i)} - \frac{(1-X_i)Y_i}{1-\widehat{\mathrm{PS}}(\boldsymbol{w}_i)}\right), \tag{10}$$

where $\widehat{\mathrm{PS}}(\boldsymbol{w}_i)$ is the estimated propensity score, defined as $\mathrm{PS}(\boldsymbol{w}_i) = P(X_i = 1 \mid \boldsymbol{w}_i)$ for observation i.

However, when the propensity scores are close to 0 or 1, estimating the average treatment effect becomes problematic with the IPW estimator. This is because the estimator becomes unstable and exhibits high variance. The truncation method [77] and the stabilizing weight method [11], [78], [79] have been proposed as solutions to this issue. Alternatively, balancing weights, including matching weights [80], overlap weights [81], [82], and entropy weights [83], have been suggested to improve efficacy and reduce imbalances when the PS model is misspecified. Nonetheless, in the presence of unmeasured confounding, their performance remains uncertain.

3) AUGMENTED INVERSE PROBABILITY OF TREATMENT WEIGHTING (AIPTW)

AIPTW is a doubly robust (DR) method that combines the properties of the regression-based estimator $\widehat{\mu}_x(\boldsymbol{w})$ and the IPTW estimator [12], [13]. The ATE can be estimated using AIPTW as

$$\begin{aligned}\widehat{\tau}_{\mathrm{AIPTW}} = {} & \frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\mu}_1(\boldsymbol{w}_i) - \widehat{\mu}_0(\boldsymbol{w}_i)\right) \\ & + \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i}{\widehat{\mathrm{PS}}(\boldsymbol{w}_i)}\left(Y_i - \widehat{\mu}_1(\boldsymbol{w}_i)\right)\right) \\ & - \frac{1}{n}\sum_{i=1}^{n}\left(\frac{(1-X_i)}{1-\widehat{\mathrm{PS}}(\boldsymbol{w}_i)}\left(Y_i - \widehat{\mu}_0(\boldsymbol{w}_i)\right)\right). \end{aligned} \tag{11}$$

AIPTW is consistent if either the $\widehat{\mu}_x(\boldsymbol{w})$ or $\widehat{\mathrm{PS}}(\boldsymbol{w})$ are consistent. In practice, only one of the two underlying models needs to be accurately specified to obtain a consistent estimator $\widehat{\tau}_{\mathrm{AIPTW}}$.

4) TARGETED MAXIMUM LIKELIHOOD ESTIMATOR (TMLE)

Like the AIPTW, the TMLE is DR. It has been shown that the TMLE and AIPTW estimators are asymptotically equivalent in many settings. However, TMLE is more robust to sparse data and near violations of practical overlap assumption than AIPTW [14]. We can estimate the ATE using TMLE as

$$\widehat{\tau}_{\mathrm{TMLE}} = \frac{1}{n}\sum_{i=1}^{n}\left(\bar{\mu}_1(\boldsymbol{w}_i) - \bar{\mu}_0(\boldsymbol{w}_i)\right), \tag{12}$$

where $\bar{\mu}_x(\boldsymbol{w}) = \widehat{\mu}_x(\boldsymbol{w}) + \widehat{\varepsilon}_1 \frac{X}{\widehat{\mathrm{PS}}(\boldsymbol{w})} + \widehat{\varepsilon}_0 \frac{1-X}{1-\widehat{\mathrm{PS}}(\boldsymbol{w})}$, $\widehat{\varepsilon}_0$ and $\widehat{\varepsilon}_1$ can be estimated using the following optimization problem as

$$\arg\min_{\varepsilon_0,\varepsilon_1} \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \widehat{\mu}_{X_i}(\boldsymbol{w}_i) + \varepsilon_1\frac{X_i}{\widehat{\mathrm{PS}}(\boldsymbol{w}_i)} + \varepsilon_0\frac{1-X_i}{1-\widehat{\mathrm{PS}}(\boldsymbol{w}_i)}\right)^2.$$

E. CAUSAL EFFECT ESTIMATION METHODS WITH UNOBSERVED CONFOUNDERS

The causal estimation methods discussed so far assume all confounding variables are observed. However, there are unobserved confounders in real observational data, which invalidates the unconfoundedness assumption and introduces bias in causal effect estimation. There is no general solution to deal with this artifact. However, in certain circumstances, one can apply approaches based on instrumental variables (IVs).

1) AN IV ESTIMATOR

IVs are another method for solving the endogeneity problem in the presence of unobserved confounders without the need to measure all adjustment confounders. The IV method relies on three key assumptions by conditioning on the baseline covariates, known as the instrumental variable assumptions: (A1) the IVs must be connected (correlated) to the treatment variable (also known as instrument relevance), (A2) the IVs have no connection with the unobserved confounders, and (A3) the IVs have no direct effect on the outcome variable, also referred to as exclusion restriction. An instrumental variable is said to be valid if it satisfies all the assumptions (A1)-(A3). In a graphical model, a valid IV is d-connected to a treatment or exposure and d-separated from an outcome when the edge of a direct effect of the treatment on the outcome is removed. In Fig. 5, Z satisfies all three assumptions which

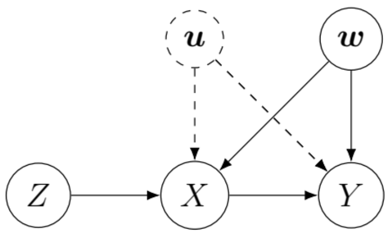

FIGURE 5. A DAG in which Z is a valid instrumental variable.

can be written using the graphical formalism given by [51] as $(Z \not\perp\!\!\!\perp X)$, $(Z \perp\!\!\!\perp \boldsymbol{u})$ and $(Z \perp\!\!\!\perp Y)_{G_{\underline{X}}})$, where $G_{\underline{X}}$ denotes the subgraph of G with all outgoing edges from nodes in X removed. Here, $X \perp\!\!\!\perp Y \mid W$ indicates that X and Y are conditionally independent given W, while $X \not\perp\!\!\!\perp Y \mid W$ indicates conditional dependence between X and Y given W. Note that (A1) is the only statistically testable assumption; the others require validation through sensitivity analyses in practice [66], [67].

In addition to the three instrumental assumptions (A1)-(A3), one needs to check the effect of homogeneity which depends on the nature of the data. Herńan and Robins [10] discussed four possible homogeneity conditions that can be considered when examining the validity of the average treatment effect. These conditions include: (a) constant effect of treatment X on outcome Y across individuals, (b) when both Z and X are binary, one can check the equality of the average causal effect of X on Y across levels of Z in both the treated and the untreated, (c) constant $X - Z$ association on the additive scale across levels of confounders $\boldsymbol{w}$, (d) monotonicity: this is an alternative homogeneity condition that does not rely on effect homogeneity.

When a binary instrumental variable Z satisfies the three instrumental assumptions (A1)-(A3) and the additional assumption of homogeneity, one can estimate the average treatment effect as

$$\tau_{IV} = \frac{(\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0])}{(\mathbb{E}[X \mid Z=1] - \mathbb{E}[X \mid Z=0])}. \tag{13}$$

For a continuous instrumental variable Z, (13) can be estimated in terms of covariance

$$\tau_{IV} = \frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(X, Z)}. \tag{14}$$

The IV estimation method differs from other methods we have discussed in the following ways: modeling assumptions are required, minor violations of the assumptions can lead to substantial biases in unpredictable directions, and it has more restrictive applicability conditions compared to alternative methods [10].

III. HIGH-DIMENSIONAL DATA: CAUSALITY CHALLENGES

High-dimensional data refers to data when the number of features or predictors p is much larger than the number of observations n. This type of dataset appears in many areas of modern science and engineering. For example, biotech data, image data, consumer preferences data, business data, and crowdsourcing data [19]. High-dimensional data has many advantages in exploring the hidden structures of each variable and identifying key common features across various data points. However, high-dimensional data present statistical complications, the most prominent being spurious correlation, endogeneity, noise accumulation, covariance estimation, locality vanishes, and computational challenges [84], [85], [86]. Hence, classical approaches developed for scenarios with large n and small p do not yield useful estimations and predictions.

High-dimensional data presents unique challenges to establish causality. Understanding causal relationships in such settings is crucial across various fields, yet it is fraught with difficulties. In the following, we provide key challenges associated with causal learning in high-dimensional scenarios.

A. SPURIOUS CORRELATION

High-dimensionality will lead to spurious correlations, which means that many unrelated random variables can have high sample correlations. Fan et al. [85] showed that when the dimensionality of the data is high, important variables can be highly correlated with several spurious variables that are scientifically unrelated. To illustrate spurious correlation in high-dimensional data, they calculated the maximum absolute sample correlation coefficient between the variable of interest and the others. Consider that $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ are n independent observations of p-dimensional Gaussian random vectors, where each $(X_1, \ldots, X_p)^\intercal \sim N(\mathbf{0}, \boldsymbol{I}_p)$. We then define

$$\widehat{r} = \max_{k \geq 2} \left| \widehat{\mathrm{Corr}}\,(X_1, X_k) \right|,$$

where $k = 2, \ldots, p$. When there is a spurious correlation among variables, the maximum absolute sample correlation increases with dimensionality. Fan et al. [87] showed that when there are many spurious variables, the variance σ^2 will be severely underestimated, leading to incorrect statistical inferences in the model selection or significance testing.

Spurious correlations can significantly impact the selection of causal variables, potentially leading to incorrect estimation of causal effect and false scientific discoveries. In high-dimensional causal analysis, the LASSO regularization method is often employed for variable selection. However, when two or more variables are highly correlated due to high-dimensionality, LASSO may select one variable over another at random, without considering the true underlying relationship. In the specific context of genomic data, where the goal is often to identify genes associated with certain traits or diseases, this random selection can lead to the inclusion of incorrect genes in the model. This poses a risk in terms of estimating causal relationships as the selected genes may not be truly causally related to the outcome of interest [85], [88].

In computer vision, processing large volumes of images or videos may result in spurious correlations due to random patterns in backgrounds [89]. These spurious correlations can adversely affect tasks such as image classification, object detection, and image generation. For instance, a model trained on images featuring a specific background pattern may incorrectly classify an object based on that pattern rather than its actual features, leading to decreased performance accuracy.

Similarly, in natural language processing (NLP), the issue of spurious correlations remains a significant challenge [90]. NLP models often rely on learning statistical correlations from large datasets to make predictions, regardless of whether these correlations reflect genuine causal relationships. While such correlations can help models predict outcomes, spurious correlations can lead to unreliable predictions by creating misleading associations that do not truly represent underlying language patterns.

Therefore, it is essential to develop a novel causal estimation method that can handle a large number of variables and effectively tackle the challenges posed by spurious correlations.

B. ENDOGENEITY

The problem of endogeneity is one of the most difficult aspects of estimating causal effects and discovering causal relationships from observational data. In contrast to spurious correlation, endogeneity refers to the incidental correlations between variables and residuals [85]. Endogeneity can significantly distort results from regression analysis, leading to unreliable and potentially misleading conclusions. For example, consider the following linear regression model

$$Y_i = \boldsymbol{x}_i^{\top} \boldsymbol{\beta} + \epsilon_i, \ i = 1, \ldots, n, \tag{15}$$

where Y_i is the ith observation of the outcome or response variable, $\boldsymbol{x}_i$ is a vector of the ith values of the predictor or explanatory variables with a dimension p much larger than n, the parameter vector $\boldsymbol{\beta}$ is sparse, which means that the majority of the components in $\boldsymbol{\beta}$ are zero and only a few predictors are important for capturing the key contributions to the regression, and ϵ_i is an unobserved random variable that adds noise to the model. A real-world application of this can be seen in genomics studies, where clinical or biological outcomes (Y_i) are often recorded along with the expression levels of thousands of genes ($\boldsymbol{x}_i$). However, in reality, only a small subset of these genes has a causal effect on the outcome.

Following the notation given by Fan and Liao [91], we assume that $\boldsymbol{\beta}_S$ represents the first s coordinates of $\boldsymbol{\beta}$ that are not vanishing. Based on this assumption, we can partition $\boldsymbol{\beta}$ as $\boldsymbol{\beta} = (\boldsymbol{\beta}_S^{\top}, \mathbf{0}_N^{\top})^{\top}$ with the corresponding predictors $\boldsymbol{x} = (\boldsymbol{x}_S^{\top}, \boldsymbol{x}_N^{\top})^{\top}$. Here, $\boldsymbol{x}_S$ represents the relevant predictors (nonzero coefficients), while $\boldsymbol{x}_N$ represents the irrelevant predictors (zero coefficients).

Predictors are called exogenous or unconfounded when $\mathbb{E}(\epsilon|\boldsymbol{x}) = 0$ and endogenous or confounded when $\mathbb{E}(\epsilon|\boldsymbol{x}) \neq 0$. The terms ''exogenous'' and ''endogenous'' are mainly used in econometrics, while ''unconfounded'' and ''confounded'' are instead used in epidemiology [92]. Thus, endogeneity undermines the foundation of regression analysis, making it impossible to establish meaningful causal inferences.

Endogeneity can arise because of several reasons, for example reverse causality or omitted variable bias [93]. However, endogeneity grows with high-dimensionality just like spurious correlations, and leads to inconsistency of classical estimation methods and false scientific conclusions. In addition, regularization techniques like the LASSO and SCAD are sensitive to endogeneity and also here can wrong sets of variables be chosen [85]. From a more general perspective, many causal methods critically depend on the exogeneity assumption. Therefore, resolving endogeneity is a vital problem in causal learning from observational data [94], [95], [96].

C. NOISE ACCUMULATION

Noise accumulation occurs when multiple parameters are estimated or tested simultaneously, resulting in estimation errors. These estimation errors accumulate when a decision or prediction rule depend on a large number of parameters.

In high-dimensional settings, where the number of parameters involved is large, noise accumulation can become particularly problematic. The presence of noise can overshadow the true signals or patterns in the data, making it hard to extract meaningful information and leading to less reliable results. Fan et al. [85] and Elman et al. [97] investigated a classification scenario with data from two classes to exemplify the problem of noise accumulation. Their result showed that high discriminative power appears possible when the number of variables is sufficiently low but decreases with increasing dimensionality. For example, in the context of genomic data, this noise can affect the classification accuracy by introducing false positive (type I error) or false negative (type II error) associations between genetic variants and traits or diseases [98]. This noise can lead to misclassification of samples, making it challenging to accurately classify individuals into different groups based on their genetic profiles. Noise accumulation is also a significant problem in computer vision for image classification.

When noise overshadows true signals or patterns, the task of extracting meaningful insights becomes challenging.

This issue can lead to unreliable causal estimates, as the underlying relationships are obscured by noise [99]. Thus, noise accumulation is especially problematic in causal discovery and causal assignment models. Causal methods must therefore incorporate denoising techniques to improve the reliability of causal conclusions, particularly in high-dimensional noisy data [100].

D. COVARIANCE ESTIMATION

Covariance estimation is a crucial tool for causal discovery, especially in high-dimensional settings where understanding conditional dependencies is essential [101]. However, accurately estimating the empirical covariance matrix in high-dimensional data poses significant challenges. Generally, the empirical covariance matrix is unreliable in high-dimensional contexts: it cannot be inverted when $p > n$ (i.e. when the number of variables exceeds the number of observations), and even if $p = n$, it may still contain errors [19], [21].

These limitations in covariance estimation pose obstacles for causal assignment models, such as principal component analysis (PCA), which relies on the covariance matrix to identify the underlying structure within the data. PCA serves as a foundational step to uncover patterns or shared dependencies useful in causal inference and controlling for confounders [102]. Similarly, causal discovery algorithms, like the PC algorithm [1], aim to uncover causal relationships by constructing causal graphs based on conditional independences and are heavily dependent on accurate covariance estimation. Errors in the estimated covariance matrix can reduce the accuracy of these algorithms, leading to incorrect or incomplete causal graphs and ultimately affecting causal inference [1], [103].

E. LOCALITY VANISHES

In high dimensions, the notion of nearest points vanishes, and any estimator based on local averaging becomes unstable [19]. For example, a causal k-nearest neighbor method can be employed to estimate the optimal treatment regime. However, this method may suffer from the curse of dimensionality, which means that its performance deteriorates as dimensionality increases [104]. Furthermore, propensity scores based on nearest-neighbor matching become increasingly biased as dimensions increase [105].

IV. CAUSAL INFERENCE IN HIGH-DIMENSIONAL SETTING WITH OBSERVED CONFOUNDERS

A. HIGH-DIMENSIONAL APPROXIMATE RESIDUAL BALANCING

High-dimensional regression adjustments (e.g., LASSO) shrink estimated effects, and neglecting this shrinkage leads to biased treatment effect estimates [106], [107]. To address this issue, Athey et al. [28] proposed the approximate residual balancing (ARB) approach, which is specifically designed for high-dimensional linear models in the potential outcomes framework. It incorporates a binary treatment indicator and all observable covariates or features to estimate treatment effects. ARB combines weight balancing with regularized regression adjustment. Specifically, the method considers the following two-step approximate residual balancing algorithm. The first stage is fitting a regularized linear model for the outcome given the features separately in the two treatment groups, e.g., using the LASSO [23], [108] or the elastic net [108]. The second stage reweights the residuals from the first-stage as proposed by Zubizarreta [109]. Residual reweighting in the second stage is linked to debiasing corrections studied in high-dimensional regression [24], [25], [26], [27]. The ARB method depends on three assumptions: unconfoundedness, sparsity, and linearity. The unconfoundedness assumption ensures that the propensity score estimation is usually consistent given that the model is correctly specified. The sparsity assumption ensures that $\boldsymbol{\beta}_0$ is s-sparse, i.e., $||\boldsymbol{\beta}_0||_0 \leq s$, where $||\boldsymbol{\beta}_0||_0$ is the ℓ_0-norm, which counts the number of nonzero elements in $\boldsymbol{\beta}_0$.

Linearity assumes that for all $\boldsymbol{w} \in \mathbb{R}^p$, the conditional response functions satisfy $\mu_0(\boldsymbol{w}) = \mathbb{E}[Y_i(0) \mid X = 0, \boldsymbol{w}_i = \boldsymbol{w}] = \boldsymbol{w}^\intercal \boldsymbol{\beta}_0$ and $\mu_1(\boldsymbol{w}) = \mathbb{E}[Y_i(1) \mid X = 1, \boldsymbol{w}_i = \boldsymbol{w}] = \boldsymbol{w}^\intercal \boldsymbol{\beta}_1$ for the control and treated units, respectively. For simplicity, the linearity assumption excludes the use of an intercept. Given this linearity, consistent estimation of treatment propensity scores is not required. Instead, residual balancing is performed. Using the linearity assumption, the problem is approached as $\tau^t = \mu_1(\boldsymbol{w}) - \mu_0(\boldsymbol{w})$, $\mu_1(\boldsymbol{w}) = \bar{\boldsymbol{w}}_1^\intercal \boldsymbol{\beta}_1$, $\mu_0(\boldsymbol{w}) = \bar{\boldsymbol{w}}_1^\intercal \beta_0$, and $\bar{\boldsymbol{w}}_1 = \frac{1}{n_1}\sum_{i=1}^{n} \mathbf{1}(\{X_i = 1\})\boldsymbol{w}_i$. Here, estimating the unbiased estimator for $\mu_0(\boldsymbol{w})$ is notably difficult when $p > n$. For example, Cassel et al. [110] and Robins et al. [111] propose to use $\widehat{\mu}_0(\boldsymbol{w}) = \bar{\boldsymbol{w}}_1\widehat{\boldsymbol{\beta}}_0 + \sum_{\{i:X_i=0\}} \tilde{\gamma}_i(Y_i - \boldsymbol{w}_i^\intercal\widehat{\boldsymbol{\beta}}_0)$. The ARB method primarily focuses on estimating $\mu_0(\boldsymbol{w})$. To solve this problem, ARB is formulated as a three-step quadratic programming problem:

1. Compute positive balancing weights $\tilde{\gamma}$ as
$$\tilde{\boldsymbol{\gamma}} = \arg\min_{\boldsymbol{\gamma}} \, (1-\zeta)\|\boldsymbol{\gamma}\|_2^2 + \zeta\,\|\bar{\boldsymbol{w}}_1 - \boldsymbol{W}_0\boldsymbol{\gamma}\|_\infty^2$$
$$\text{subject to} \sum_{\{i:X_i=0\}} \gamma_i = 1 \text{ and } 0 \leq \gamma_i \leq n_0^{\frac{-2}{3}}, \quad (16)$$
where $\boldsymbol{W}_0$ represents the feature matrix for the control units.
2. Obtain estimates of $\boldsymbol{\beta}_0$ using the elastic net (or LASSO) as
$$\widehat{\boldsymbol{\beta}}_0 = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{\{i:X_i=0\}} \left(Y_i - \boldsymbol{w}_i^\intercal\boldsymbol{\beta}\right)^2 + \lambda\left\{(1-\alpha)\|\boldsymbol{\beta}\|_2^2 + \alpha\|\boldsymbol{\beta}\|_1\right\}\right\}. \quad (17)$$
3. Estimate the ATT as
$$\widehat{\tau}^t_{\text{ARB}} = \frac{1}{n_1}\sum_{\{i:X_i=1\}} Y_i - \left\{\bar{\boldsymbol{w}}_1^\intercal\widehat{\boldsymbol{\beta}}_0 + \sum_{\{i:X_i=0\}} \tilde{\gamma}_i\left(Y_i - \boldsymbol{w}_i^\intercal\widehat{\boldsymbol{\beta}}_0\right)\right\}, \quad (18)$$

where $\zeta \in (0,1)$, $\alpha \in (0,1]$ and $\lambda > 0$ are tuning parameters.

The choice of parameter ζ in (16) is a problem because we are trying to balance sampling variance against residual bias [28]. It is not possible to use cross-validation since (16) is a regularized and constrained unsupervised optimization problem. Data-adaptive choices for ζ would be of considerable interest. One can also estimate ATE as

$$\hat{\tau}_{\text{ARB}} = \bar{\boldsymbol{w}}^{\top}\left(\widehat{\boldsymbol{\beta}}_1 - \widehat{\boldsymbol{\beta}}_0\right) + \sum_{i:X_i=1} \tilde{\gamma}_{1,i}\left(Y_i - \boldsymbol{w}_i^{\top}\widehat{\boldsymbol{\beta}}_1\right) - \sum_{i:X_i=0} \tilde{\boldsymbol{\gamma}}_{0,i}\left(Y_i - \boldsymbol{w}_i^{\top}\widehat{\boldsymbol{\beta}}_0\right), \tag{19}$$

where

$$\tilde{\boldsymbol{\gamma}}_1 = \arg\min_{\boldsymbol{\gamma}} (1-\zeta)\|\boldsymbol{\gamma}\|_2^2 + \zeta\|\bar{\boldsymbol{w}} - \boldsymbol{W}_1\boldsymbol{\gamma}\|_\infty^2$$
$$\text{subject to} \sum_{\{i:X_i=1\}} \gamma_i = 1 \text{ and } 0 \le \gamma_i \le n_1^{\frac{-2}{3}}. \tag{20}$$

$\tilde{\boldsymbol{\gamma}}_0$ is defined analogously.

Here, $\boldsymbol{W}_1$ represents the feature matrix for the treated units. This strategy can be extended to generalized linear models, where ψ is a nonlinear link function for which $\mathbb{E}[Y_i(0) \mid \boldsymbol{w}_i] = \psi(\boldsymbol{w}_i^{\top}\boldsymbol{\beta}_0)$. In the context of DR estimation, Cassel et al. [110] and Robins et al. [111] proposed an augmented inverse propensity weighting (AIPTW) estimator given some estimated propensity scores $\widehat{\text{PS}}(\boldsymbol{w}_i)$ as

$$\widehat{\mu_0}_{\text{AIPTW}} = \bar{\boldsymbol{w}}_1^{\top}\widehat{\boldsymbol{\beta}}_0 + \frac{\sum_{i:X_i=1} \frac{\widehat{\text{PS}}(\boldsymbol{w}_i)}{1-\widehat{\text{PS}}(\boldsymbol{w}_i)}\left(Y_i - \boldsymbol{w}_i^{\top}\widehat{\boldsymbol{\beta}}_0\right)}{\sum_{i:X_i=0} \frac{\widehat{\text{PS}}(\boldsymbol{w}_i)}{1-\widehat{\text{PS}}(\boldsymbol{w}_i)}}. \tag{21}$$

Equation (21) is DR in the sense that it is consistent if either the propensity fit $\widehat{\text{PS}}(\boldsymbol{w}_i)$ or the outcome fit $\widehat{\beta}_0$ is consistent [74], [112], [113]. However, when $1 - \hat{\text{PS}}(\boldsymbol{w}_i)$ is near 0, this class of methods may perform poorly [74], [75]. To solve this issue, one can use the truncation method [77] or the stabilizing weight method [11], [78] as discussed under the IPTW method.

B. BAYESIAN ADDITIVE REGRESSION TREES FOR ESTIMATING CAUSAL EFFECTS

Hill [114] proposed a Bayesian additive regression trees (BART) procedure for flexible modeling of the outcome variable to estimate the treatment effect.

Under this framework, Hill [114] considers the case where X_i is a binary treatment indicator such that $X_i = 1$ when the treatment is given to the ith individual, and $X_i = 0$ otherwise, with corresponding potential outcomes $Y_i(1)$ and $Y_i(0)$, respectively, along with the observable vector of covariates $\boldsymbol{w}_i$. The implementation of BART requires the treatment assignment and covariates to model the outcome and does not require information about how these variables are parametrically related. One can model the outcome as

$$Y = f(x, \boldsymbol{w}) + \epsilon \text{ with } f(x, \boldsymbol{w}) = \sum_{j=1}^{K} g(x, \boldsymbol{w}, T_j, M_j), \tag{22}$$

where each (T_j, M_j) denotes a single subtree model, $M_j = \{\mu_{j1}, \mu_{j2}, \ldots, \mu_{jb}\}$ contains a set of bottom nodes, b is the number of bottom nodes, μ_{jb} is the mean outcome of the bth node of the jth subtree classified to this node, $\epsilon \sim N(0, \sigma^2)$ and K is the number of regression trees. Assuming that additive errors and unconfoundedness hold conditional on $\boldsymbol{w}$, one can estimate the CATE as

$$\tau(\boldsymbol{w}) = \frac{1}{n}\sum_{i=1}^{n} \left(\mathbb{E}[Y_i(1) \mid \boldsymbol{w}_i = \boldsymbol{w}] - \mathbb{E}[Y_i(0) \mid \boldsymbol{w}_i = \boldsymbol{w}]\right)$$
$$= \frac{1}{n}\sum_{i=1}^{n} \left(f(1, \boldsymbol{w}_i) - f(0, \boldsymbol{w}_i)\right), \tag{23}$$

where $f(x, \boldsymbol{w}) = \mathbb{E}[Y \mid X = x, \boldsymbol{w}]$. Comparisons have shown that BART outperforms linear regression, propensity score matching, and an IPTW estimator [114]. In comparison to other approaches like random forests, boosting and neural networks, BART has some advantages [114], [115]. Some examples include easier tuning of hyperparameters, quantification of uncertainty by inferring the posterior distribution of outcomes and efficient handling very large numbers of continuous and discrete variables. Hill [114] suggested the use of a regularization prior that constrains the size and fit of each tree to directly estimate $f(x, \boldsymbol{w}_i)$. However, Hahn et al. [116] instead recommend expressing the outcome as $\mathbb{E}[Y_i \mid X_i, \boldsymbol{w}_i] = \Delta(\boldsymbol{w}_i, \text{PS}(\boldsymbol{w}_i)) + \pi(\boldsymbol{w}_i)X_i$, where the treatment effect $\pi(\boldsymbol{w}_i) = f(1, \boldsymbol{w}_i) - f(0, \boldsymbol{w}_i)$ can be regularized independent of $\Delta(\cdot)$ using a separate prior distribution. Incorporating an estimation of the propensity function in this context serves to avoid biased estimates of treatment effects when fitting the model to data that contains strong confounding.

C. DOUBLE MACHINE LEARNING METHOD

Both SCMs and PO frameworks rely on stringent conditional independence assumptions, often difficult to meet when not all confounders are measured. In contrast, econometrics bases causal identification on moment restrictions, providing a more flexible approach by focusing on error terms instead of requiring full conditional independence. These restrictions are tractable; functional form assumptions allow relaxation of conditional independence to conditional mean independence and, in some cases, to orthogonality of error terms.

Chernozhukov et al. [107] present a double machine learning (DML) method that applies various functionalities to estimate causal effects in different causal models. Among these models, they propose estimating causal effects using a partially linear regression (PLR) model based on [117], which is specified following

$$X = g(\boldsymbol{w}) + U_X, \ \mathbb{E}(U_X \mid \boldsymbol{w}) = 0, \tag{24}$$
$$Y = X\theta + f(\boldsymbol{w}) + U_Y, \ \mathbb{E}(U_Y \mid X, \boldsymbol{w}) = 0, \tag{25}$$

where Y is the outcome variable, X is the policy (treatment) variable of interest, and $\boldsymbol{w}$ represents the high-dimensional vector of covariates. Here, (24) accounts for confounding, i.e., the dependence of the treatment variable on the covariate $\boldsymbol{w}$, (25) is the main equation, θ is a scalar parameter of interest, U_X and U_Y are random noise terms. If X is unconfounded conditioned on the covariate $\boldsymbol{w}$, then the parameter θ is interpreted as the treatment effect. The confounding factors $\boldsymbol{w}$ affect the treatment variable X through the function $g(\boldsymbol{w})$ and the outcome variable through the function $f(\boldsymbol{w})$. The nuisance parameter $\kappa = \kappa(g, f)$ denotes the high-dimensional nuisance functions g and f, which are estimated using machine learning methods.

1) REGULARIZATION BIAS

One potential method to estimate the parameter θ involves the employment of a machine learning estimator $X\widehat{\theta}+\widehat{f}(\boldsymbol{w})$ to learn the outcome function $Y = X\theta+f(\boldsymbol{w})$. If one is confident that f can be adequately approximated by a sparse linear combination of pre-specified functions of $\boldsymbol{w}$, one may opt to use the LASSO. In alternative situations, one could consider using iterative techniques that involve alternating between random forests and least squares to estimate θ. To avoid overfitting, perform data splitting by randomly dividing the data into two parts, each with sizes n_1 and n_2, respectively. Let I_1 be the set of indices in n_1, and I_2 be the set of indices in n_2. Assuming $\widehat{f}$ is obtained using the data in I_2, then the final estimate of θ is obtained using the data in I_1 as

$$\widehat{\theta} = \left(\frac{1}{n_1}\sum_{i\in I_1} X_i^2\right)^{-1} \frac{1}{n_1}\sum_{i\in I_1}(Y_i - \widehat{f}(\boldsymbol{w}_i)). \tag{26}$$

Chernozhukov et al. [107] demonstrated that the regularization method induces substantive biases in the estimation of $\widehat{f}$.

2) OVERCOMING REGULARIZATION BIASES USING ORTHOGONALIZATION

This method employs a construction approach that utilizes an orthogonalized formulation achieved by directly partialling out the effect of $\boldsymbol{w}$ from X. Thus, a new orthogonalized residual is formed and estimated as $\widehat{U}_X = X - \widehat{g}(\boldsymbol{w})$, where $\widehat{g}$ is a machine learning estimator of g. Machine learning methods include random forests, lasso or post-lasso, neural nets, boosted regression trees, as well as several hybrid and ensemble models that combine these methods. One can solve a prediction problem to estimate the conditional mean of X given $\boldsymbol{w}$, i.e. doing double prediction or double machine learning. After partialling out the effect of $\boldsymbol{w}$ from X and obtaining an estimate of f using the data in I_2, the double or debiased machine learning (DML) estimator for θ can be obtained using the data in I_1 according to

$$\widehat{\theta}_{\text{DML}} = \left(\frac{1}{n_1}\sum_{i\in I_1} \widehat{U}_{X_i}X_i\right)^{-1} \left(\frac{1}{n_1}\sum_{i\in I_1} \widehat{U}_{X_i}\left(Y_i - \widehat{f}(\boldsymbol{w}_i)\right)\right). \tag{27}$$

One can observe that (27) eliminates regularization bias in (26) by approximating orthogonalization of X with respect to $\boldsymbol{w}$ and subtracting an estimate of f to remove the direct effect of confounding.

Chernozhukov et al. [107] utilized DML to estimate the ATE and the ATT by constructing Neyman-orthogonal scores [118]. For example, in the setting of an interactive regression model (IRM), the data are represented as vectors $(\boldsymbol{y}, \boldsymbol{x}, \boldsymbol{w})$, where

$$\begin{aligned} \boldsymbol{x} &= g(\boldsymbol{w}) + U_{\boldsymbol{x}}, \quad \mathbb{E}(U_{\boldsymbol{x}} \mid \boldsymbol{w}) = 0,\\ \boldsymbol{y} &= f(\boldsymbol{x}, \boldsymbol{w}) + U_{\boldsymbol{y}}, \quad \mathbb{E}(U_{\boldsymbol{y}} \mid \boldsymbol{x}, \boldsymbol{w}) = 0,\\ \tau_{\text{DML}} &= \mathbb{E}\left[f(1, \boldsymbol{w}) - f(0, \boldsymbol{w})\right],\\ \tau_{\text{DML}}^{t} &= \mathbb{E}\left[f(1, \boldsymbol{w}) - f(0, \boldsymbol{w}) \mid \boldsymbol{x} = 1\right]. \end{aligned}$$

Here, the treatment variable $\boldsymbol{x}$ is binary ($\boldsymbol{x} \in \{0, 1\}$), $f(\boldsymbol{x}, \boldsymbol{w}) = \mathbb{E}[\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}]$, and $g(\boldsymbol{w}) = \text{PS}(\boldsymbol{w})$. In addition to PLR and IRM, the DML method also provides detailed algorithms for the estimation of causal effects in partially linear instrumental variable regression models (PLIV) and interactive instrumental variable regression models (IIVM) [119].

V. CAUSAL INFERENCE IN HIGH-DIMENSIONAL SETTING WITH UNOBSERVED CONFOUNDERS

A. HIGH-DIMENSIONAL TWO-STAGE LEAST SQUARES (2SLS) ESTIMATION

The practical implementation of 2SLS revolves around the identification of a valid instrument variable (IV). The first stage involves estimating the extent to which modifying the relevant IV would affect a particular treatment. In the second stage, one examines how the changes in the treatment caused by the IV would affect the outcome.

Consider a high-dimensional dataset $(\boldsymbol{y}, \boldsymbol{X}, \boldsymbol{Z})$, where $\boldsymbol{y}$ is the outcome vector of dimension $n \times 1$, $\boldsymbol{X}$ is the $n \times p$ matrix of treatment variables, and $\boldsymbol{Z}$ is the $n \times q$ matrix of IVs, corresponding to the causal model shown in Fig. 6. Here, the only common causes of $\boldsymbol{X}$ and $\boldsymbol{y}$ are the unobserved confounders $\boldsymbol{U}$. The structural linear equations compatible with Fig. 6 can be written as

$$\begin{aligned} \boldsymbol{X} &= \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\eta},\\ \boldsymbol{y} &= \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \end{aligned} \tag{28}$$

where $\boldsymbol{\beta}$ is a $p \times 1$ vector that represents the direct effects of the components of $\boldsymbol{X}$ on $\boldsymbol{y}$, $\boldsymbol{\gamma}$ is a $q \times p$ matrix that represents the direct effects of the components of $\boldsymbol{Z}$ on $\boldsymbol{X}$, $\boldsymbol{\epsilon} = \boldsymbol{U} + \boldsymbol{U_y}$

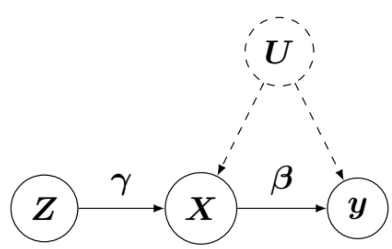

FIGURE 6. A DAG emphasizing the estimation of β using the instrumental variable method.

is an $n \times 1$ vector of noise terms, and $\boldsymbol{\eta} = \boldsymbol{U} + \boldsymbol{U}_X$ is an $n \times p$ matrix of noise terms.

The aim is to estimate the causal parameter of interest $\boldsymbol{\beta}$ which represents the direct effect of the components of the causal variable $\boldsymbol{X}$ on the outcome variable $\boldsymbol{y}$. However, here the direct effect of $\boldsymbol{X}$ on $\boldsymbol{y}$ cannot be identified through adjustment due to the unobserved common cause $\boldsymbol{U}$.

In this case, the IV variable $\boldsymbol{Z}$ can be used to identify the direct effect. Since the optimal instruments are unknown, it is assumed that the number of instruments is at least equal to the number of endogenous $\boldsymbol{X}$ (those correlated with the error term). Assuming model (28) is sparse, i.e. only a small portion of the coefficients in $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ is nonzero. As a result, the goal is to estimate the nonzero coefficients in both $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$. Application of one-stage regularization without the use of instruments leads to the penalized least squares estimator which is not a consistent estimator of $\boldsymbol{\beta}$. As a result, Lin et al. [120] proposed a two-stage regularization (2SR) method using regularization methods in both stages of the 2SLS method to deal with high-dimensionality. The first stage is to identify and estimate the nonzero coefficients of the instruments, as well as to obtain the predicted values of the causal variables. The regularized estimator for the first stage is defined as

$$\hat{\boldsymbol{\gamma}} = \underset{\boldsymbol{\gamma} \in \mathbb{R}^{q \times p}}{\arg\min} \left\{ \frac{1}{2n} \|\boldsymbol{X} - \boldsymbol{Z}\boldsymbol{\gamma}\|_F^2 + \sum_{i=1}^{q} \sum_{j=1}^{p} P_{\lambda_j}(\gamma_{ij}) \right\}, \quad (29)$$

where $\|.\|_F$ denote the Frobenius norm, γ_{ij} is the $(i, j)th$ entry of $\boldsymbol{\gamma}$, P_{λ_j} is a sparsity-inducing penalty function that will be discussed later, and $\lambda_j > 0$ are tuning parameters that control the strength of the first-stage regularization. $\hat{\boldsymbol{X}} = \boldsymbol{Z}\hat{\boldsymbol{\gamma}}$ forms the predicted value of $\boldsymbol{X}$ after the estimate is obtained. The second stage amounts to identify and estimate the nonzero coefficients of the causal variables by substituting the first-stage prediction $\hat{\boldsymbol{X}}$ for $\boldsymbol{X}$. The regularized estimator for the second stage is specified as

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta} \in \mathbb{R}^{p}}{\arg\min} \left\{ \frac{1}{2n} \|\boldsymbol{y} - \widehat{\boldsymbol{X}}\boldsymbol{\beta}\|_2^2 + \sum_{j=1}^{p} P_{\nu}(\beta_j) \right\}, \quad (30)$$

where $\nu > 0$ is a tuning parameter that controls the regularization in the second stage. As a result, we get the pair $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$ as the final estimator of the coefficients $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ in the model (28).

As penalization terms, one can choose a LASSO penalty function $P_\lambda(t) = \lambda|t|$ [23], [121]. One may consider employing alternative penalty functions, e.g. SCAD [122] and MCP [123]. The coordinate descent algorithm can be implemented to solve the optimization problems in (29) and (30) Lin et al. [120].

B. HIGH-DIMENSIONAL OVER-IDENTIFICATION

In this section, we consider the scenario of high-dimensional over-identification, where the linear outcome model is given by (15) with particular attention to estimating the true causal effect β_j of the variable x_{ij} on Y_i, where $j = 1, \dots, p$ and $i = 1, \dots, n$.

Assume that the only common causes of $\boldsymbol{x}_i$ and Y_i are the unknown confounders $\boldsymbol{u}_i$ (as shown in Fig. 6), and that the majority of the components in $\boldsymbol{\beta}$ are zero, meaning that only a few variables in $\boldsymbol{x}_i$ cause the outcome variable Y_i. In this scenario, one can use moment restrictions as an alternative to SCMs and PO, specifically the generalized method of moments (GMM). In particular, one can leverage the property of valid instrumental variables and express it as a moment condition: $\mathbb{E}(\epsilon_i z_i) = 0$, where z_i denotes a q-dimensional vector of instrumental variables. Consequently, the moment condition using (15) is specified as

$$\mathbb{E}\left[(Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) z_i\right] = \mathbb{E}\left[m(\boldsymbol{x}_i, Y_i, z_i, \boldsymbol{\beta})\right] = 0, \quad (31)$$

where $m(\boldsymbol{x}_i, Y_i, z_i, \boldsymbol{\beta}) = (Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) z_i$. Our interest lies in the case where the dimension of m is larger than the dimension of the true parameter $\boldsymbol{\beta}$ (i.e., $q > p$).

Our interest is when the dimension of m is larger than the dimension of the true parameter $\boldsymbol{\beta}$ (i.e. $q > p$). This is referred to as *over-identification*, which means that the number of moment conditions (number of equations) is larger than the number of unknown parameters. If a model is over-identified, then a solution to (31) is unlikely to exist. To deal with this issue, another GMM is obtained by using a weight matrix in the quadratic form to estimate $\boldsymbol{\beta}$ as

$$\hat{\boldsymbol{\beta}}_{\text{GMM}} = \underset{\boldsymbol{\beta}}{\arg\min} \left[\sum_{i=1}^{n} (Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) Z_i \right]^\top \boldsymbol{J} \left[\sum_{i=1}^{n} (Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) Z_i \right], \quad (32)$$

where $\boldsymbol{J}$ is a $q \times q$ positive definite weighting matrix computed from available data [124].

We now consider the high-dimensional $\boldsymbol{x}_i$, where only some of its components affect Y_i. To accommodate this, we partition $\boldsymbol{\beta}$ as $\boldsymbol{\beta} = (\boldsymbol{\beta}_S^\top, \boldsymbol{\beta}_N^\top)^\top$ and the corresponding variable $\boldsymbol{x}$ as $\boldsymbol{x} = (\boldsymbol{x}_S^\top, \boldsymbol{x}_N^\top)^\top$. Here, $\boldsymbol{x}_S$ refers to the vector of causal variables $\boldsymbol{x}$ with dimension S, and $\boldsymbol{x}_N$ refers to the vector of non-causal variables $\boldsymbol{x}$ with dimension N.

In reality, the dimension of $\boldsymbol{x}_N$ is large, and hence, some of its components can affect Y through unobserved confounders, i.e., $\mathbb{E}(\epsilon \boldsymbol{x}_N) \neq 0$. Therefore, the working model for a single observation is $Y = \boldsymbol{x}_S^\top \boldsymbol{\beta}_S + \boldsymbol{x}_N^\top \boldsymbol{\beta}_N + \epsilon$. However, once $\boldsymbol{x}_S$ is identified, the true model can be written as

$$Y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon = \boldsymbol{x}_S^\top \boldsymbol{\beta}_S + \epsilon, \ \mathbb{E}(Y - \boldsymbol{x}_S^\top \boldsymbol{\beta}_S | \boldsymbol{x}_S) = 0. \quad (33)$$

The concept of over-identification allows a model to be identified. For any $R \subset S$ with size $|R|$ and $S \in \{1, \dots, p\}$, there exists a solution to the over-identified equations following

$$\mathbb{E}(Y - \boldsymbol{x}_R^\top \boldsymbol{\beta}_R) \boldsymbol{x}_R = 0, \text{ and } \mathbb{E}(Y - \boldsymbol{x}_R^\top \boldsymbol{\beta}_R) \boldsymbol{x}_R^2 = 0, \quad (34)$$

where $\boldsymbol{x}_R^2$ is a vector consisting of the squared components of $\boldsymbol{x}_R$ (it can be replaced with other nonlinear functions, for example, $|\boldsymbol{x}_R|$). Equation (34) yields a system of equations that is twice the number of unknown parameters. Nonetheless the solution exists and is given by $\boldsymbol{\beta}_R = \boldsymbol{\beta}_S$ since $\boldsymbol{\beta}_S$ validates more equations than its dimension. On the other hand, if $R \not\subset S$, (34) has no solution since there are more equations than the number of unknown parameters. Fan and Liao [91] introduced a more general loss function called the focused generalized method of moments (FGMM) to estimate $\boldsymbol{\beta}_S$. Unlike the low-dimensional techniques of either moment selection [125], [126] or shrinkage GMM [127], the FGMM method can be used to detect incorrectly specified moment conditions in the form of $\mathbb{E}\left[(Y - \boldsymbol{x}_S^\top \boldsymbol{\beta}_S)\boldsymbol{x}_N\right] \neq 0$ if $\boldsymbol{x}_N$ is an endogenous variable. The loss function for FGMM is

$$\begin{aligned} L_{\mathrm{FGMM}(\boldsymbol{\beta})} &= \sum_{j=1}^{p} \boldsymbol{I}_{(\beta_j \neq 0)} \left[\frac{1}{\widehat{\mathrm{var}}(x_j)} \left(\sum_{i=1}^{n} g(Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) x_{ij} \right)^2 \right. \\ &\quad \left. + \frac{1}{\widehat{\mathrm{var}}(x_j^2)} \left(\sum_{i=1}^{n} g(Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) x_{ij}^2 \right)^2 \right] \\ &= \left[\sum_{i=1}^{n} g(Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) z_i(\boldsymbol{\beta}) \right]^\top \\ &\quad \times \boldsymbol{J}(\boldsymbol{\beta}) \left[\sum_{i=1}^{n} g(Y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}) z_i(\boldsymbol{\beta}) \right], \end{aligned} \tag{35}$$

where $\boldsymbol{J}(\boldsymbol{\beta})$ is the diagonal weight matrix computed from the given data, $\widehat{\mathrm{var}}$ represents the sample variance, $z_i(\boldsymbol{\beta}) = (\boldsymbol{x}_i(\boldsymbol{\beta})^\top, \boldsymbol{x}_i^2(\boldsymbol{\beta})^\top)^\top$ is the instrumental variable containing the $|\boldsymbol{\beta}|$-dimensional vectors $\boldsymbol{x}_i(\boldsymbol{\beta})$ and $\boldsymbol{x}_i^2(\boldsymbol{\beta})$. The indicator functions $\boldsymbol{I}_{(\beta_j \neq 0)}$ in the loss function are used to handle the problems of endogeneity and model misspecification simultaneously. Because of this, the instrumental variable $z(\boldsymbol{\beta})$ is dependent on the parameter $\boldsymbol{\beta}$ which leads to the focussed GMM. However, the indicator function is insufficient to ensure consistency in variable selection. Therefore, a LASSO penalty function $\|P_n(\boldsymbol{\beta})\|_1 = \sum_{j=1}^{p} P_n(|\beta_j|)$ is required to set the unimportant $\boldsymbol{x}$ to zero. Hence, the FGMM criterion function can be written as $L_{\mathrm{FGMM}}(\boldsymbol{\beta}) + \|P_n(\boldsymbol{\beta})\|_1$. Minimizing the criteria function yields the FGMM estimator

$$\widehat{\boldsymbol{\beta}}_{\mathrm{FGMM}} = \underset{\boldsymbol{\beta}}{\arg\min} \left\{ L_{\mathrm{FGMM}}(\boldsymbol{\beta}) + \|P_n(\boldsymbol{\beta})\|_1 \right\}. \tag{36}$$

As discussed earlier, adding an indicator function to the loss function is useful for dimension reduction and handling endogeneity. Unfortunately, it also makes L_{FGMM} non-smooth. As a result, the criterion function is generally not continuous, and minimizing it without further consideration is typically NP-hard, as no algorithms exist to solve this problem in polynomial time. However, by using a smoothing technique that approximates the indicator function with a smooth kernel function [91], [128], [129], one can solve the discontinuity problem. Furthermore, in the loss function, both X and X^2 are needed for over-identification. To numerically minimize function (36), one can use the coordinate descent algorithm with a kernel-smoothing function [91].

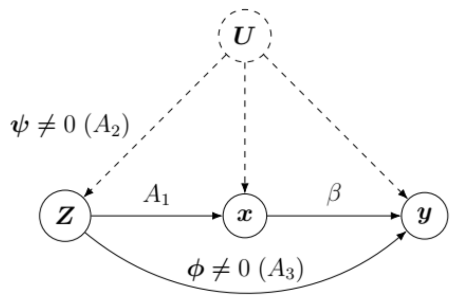

FIGURE 7. DAG illustration of a possible violation of IV assumptions A_2 and A_3.

C. HIGH-DIMENSIONAL IV ESTIMATION WITH SOME INVALID INSTRUMENTS

Traditional estimation methods using IV require complete knowledge of the validity of all instruments. This is often infeasible, as have been demonstrated by Mendelian randomization studies in which genetic markers are used as instruments [130]. Complete knowledge of instrument validity is equivalent to complete knowledge of the biochemical function of all involved genes. To solve this problem, Kang et al. [131] proposed a method to estimate causal effects without complete knowledge of the validity of all instrumental variables. Specifically, the proposed method allows for possible violation of either assumption (A2) or (A3) in estimating causal effects. Fig. 7 shows a graphical model for this problem. Here, $\boldsymbol{y} \in \mathbb{R}^n$ is a vector of observed outcomes, $\boldsymbol{x} \in \mathbb{R}^p$ is a vector of observed treatments, $\boldsymbol{Z}$ denotes a $\mathbb{R}^{n \times q}$ matrix of instruments, where row i consists of z_i, and $\boldsymbol{U}$ denotes unobserved confounders. The parameter $\boldsymbol{\phi}$ represents the direct effect of the instruments on the outcome (also called the pleiotropic effect in genetics [92]). Likewise, the parameter $\boldsymbol{\psi}$ represents the effect confounders $\boldsymbol{U}$ that affect the instrument. The parameter β represents the scalar causal parameter of interest, which is the constant effect of the treatment on the outcome. We now have the following observed outcome model that after some re-arrangement becomes

$$\boldsymbol{y} = \boldsymbol{Z}\boldsymbol{\alpha} + \boldsymbol{x}\beta + \boldsymbol{\epsilon}, \ \mathbb{E}(\boldsymbol{\epsilon} \mid \boldsymbol{Z}) = 0), \tag{37}$$

where $\boldsymbol{\alpha} = \boldsymbol{\phi} + \boldsymbol{\psi}$. The IV is valid if $\alpha_j = 0$ and invalid if $\alpha_j \neq 0$ for $j \in \{1, \ldots, q\}$. The parameters $\boldsymbol{\alpha}$ and β can be estimated by using L_1 penalization as

$$\widehat{\boldsymbol{\alpha}_\lambda}, \ \widehat{\beta_\lambda} \in \underset{\boldsymbol{\alpha},\, \beta}{\arg\min} \|\boldsymbol{P_Z}(\boldsymbol{Z}^\top(\boldsymbol{y} - \boldsymbol{Z}\boldsymbol{\alpha} - \boldsymbol{x}\beta))\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_1, \tag{38}$$

where $\boldsymbol{P_Z} = \boldsymbol{Z}(\boldsymbol{Z}^\top\boldsymbol{Z})^{-1}\boldsymbol{Z}^\top$ is the projection onto the column space of $\boldsymbol{Z}$, $\|\boldsymbol{\alpha}\|_1 = \sum_j |\alpha_j|$ and $\lambda > 0$ is a tuning parameter. This problem is solvable depending on the following two assumptions: (a) $\mathbb{E}(\boldsymbol{Z}^\top\boldsymbol{Z})$ is a full rank, and (b) $\mathbb{E}(\boldsymbol{Z}^\top\boldsymbol{x}) = \mathbb{E}(\boldsymbol{Z}^\top(\boldsymbol{Z})\boldsymbol{\gamma}^*)$, the components of $\boldsymbol{\gamma}^*$ are all not equal to zero, i.e., $\gamma_j^* \neq 0$. Assumption (a) specifies that the instrument $\boldsymbol{Z}$ matrix is full rank, which is a typical

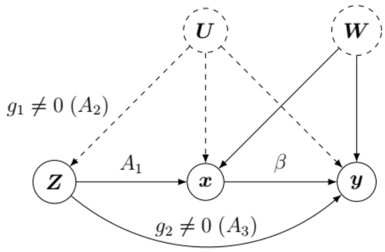

FIGURE 8. A graphical model extending Fig. 7 with an observed confounder matrix W.

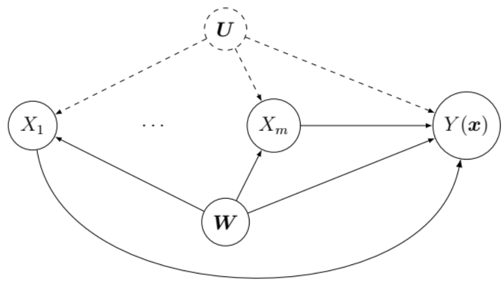

FIGURE 9. DAG setup for deconfounder method.

assumption in the literature on instrumental variables [132]. Assumption (b) states that the instruments are relevant to the treatment, as in assumption (A1). One can solve (38) using an R-function called sisVIVE (some invalid and some valid IV estimator), where the regularization parameter λ can be obtained by cross-validation (CV) [131]). Guo and Bühlmann [133] extended (37) by incorporating a matrix of observed confounders W and proposing the two stage curvature identification (2SCI) method. However, instead of considering a path coefficient, the estimation in this scenario is based on the function g. In this context, the function g will be called the violation function, and a nonzero value of g indicates a violation of assumptions (A2) and (A3). Fig. 8 illustrates the graphical model for this scenario. They consider i.i.d. data $(\boldsymbol{y}, \boldsymbol{x}, \boldsymbol{Z}, \boldsymbol{W})$ and the outcome model as

$$\boldsymbol{y} = \boldsymbol{x}\beta + g(\boldsymbol{W}, \boldsymbol{Z}) + \boldsymbol{\gamma} \text{ with } \mathbb{E}(\boldsymbol{\gamma} \mid \boldsymbol{W}, \boldsymbol{Z}) = 0, \tag{39}$$

where $g(\cdot) = g_1(\cdot) + g_2(\cdot)$. For simplicity, the treatment-outcome model without confounders can be formulated as

$$\begin{aligned} \boldsymbol{x} &= f(\boldsymbol{Z}) + \boldsymbol{\sigma}, \\ \boldsymbol{y} &= F(\boldsymbol{Z}) + \boldsymbol{\gamma} + \beta\boldsymbol{\sigma} \text{ with } F(\boldsymbol{Z}) = \beta f(\boldsymbol{Z}) + g(\boldsymbol{Z}). \end{aligned} \tag{40}$$

In the presence of a confounder $\boldsymbol{W}$, (40) can be written as

$$\begin{aligned} \boldsymbol{x} &= f(\boldsymbol{Z}, \boldsymbol{W}) + \boldsymbol{\sigma}, \\ \boldsymbol{y} &= F(\boldsymbol{Z}, \boldsymbol{W}) + \boldsymbol{\gamma} + \beta\boldsymbol{\sigma}, \\ \text{where } F(\boldsymbol{Z}, \boldsymbol{W}) &= \beta f(\boldsymbol{Z}, \boldsymbol{W}) + \boldsymbol{\omega}(\boldsymbol{W}) + g(\boldsymbol{Z}, \boldsymbol{W}). \end{aligned} \tag{41}$$

A key problem here is to estimate $f(\boldsymbol{Z}, \boldsymbol{W}) = \mathbb{E}(\boldsymbol{x} \mid \boldsymbol{W}, \boldsymbol{Z})$. Guo and Bühlmann [133] developed 2SCI with an estimation of the conditional mean function by random forests, with extensions to general machine learning methods like boosting, deep neural networks and basis function approximation. The first stage is to estimate $f(\boldsymbol{Z}, \boldsymbol{W})$ by random forests. The second stage is to employ the estimator obtained from the first stage and make necessary adjustments for the violations.

D. DECONFOUNDER ALGORITHM FOR MULTIPLE CAUSES

Wang and Blei [102] developed the deconfounder algorithm which describes a way to estimate causal effects in multiple-cause settings with weaker assumptions than traditional methods require. There are m possible causes or treatments $\boldsymbol{x} = (x_1, \ldots, x_m)^\intercal$ and the outcome $Y(\boldsymbol{x})$. These causes can take the form of real values, binary values, or integers. In this scenario, assume that patients receive various treatments and that the outcome will depend on the specific treatment received. Here, for each individual i, there are m different assigned treatments $\boldsymbol{x_i} = (x_{i1}, \ldots, x_{im})^\intercal$ and the outcome $Y_i(\boldsymbol{x})$. As discussed in previous sections, when the covariates $\boldsymbol{W}$ contain all confounders and block back-door paths (i.e., $\boldsymbol{W}$ also satisfies the unconfoundedness assumption), then the following holds

$$\mathbb{E}\left[Y_i(\boldsymbol{x})\right] = \mathbb{E}\left[\mathbb{E}\left[Y_i(\boldsymbol{x}) \mid W_i, \boldsymbol{X}_i = \boldsymbol{x}\right]\right]. \tag{42}$$

Here, the core idea is to construct and infer a random variable U_i that can serve as a substitute confounder in the analysis, rather than measuring potential confounders. The random variable U_i is a multi-cause separator that renders all causes $X_{i1}, \ldots, X_{im}$ conditionally independent.

The deconfounder algorithm further depends on the "no unobserved single-cause" and "weak unconfoundedness" assumptions. No unobserved single-cause assumption requires that one can observe any confounders that affect only one of the causes. In Fig. 9, the deconfounder algorithm requires that $\boldsymbol{U}$ contains all unobserved multi-cause separators. The weak unconfoundedness assumption states that the individual potential outcomes $Y_i(\boldsymbol{x})$ are marginally independent of the causes $\boldsymbol{X}_i$ (i.e. $X_{i1}, \ldots, X_{im} \perp\!\!\!\perp Y_i(\boldsymbol{x}) \mid U_i, W_i$ for all $\boldsymbol{x}$) [134]. This means that no unobserved single cause requires all unobserved confounders to be multi-cause variables that influence at least two causal treatment variables, and that, together with the observed confounders, they satisfy the weak unconfoundedness assumption. Furthermore, the deconfounder method assumes that there are no causal arrows between the treatment variables (i.e., the causes are not causally dependent).

If the estimation of the substitute confounder $\boldsymbol{U}$ is successfully obtained, the observed confounders should be replaced with $\boldsymbol{W}$ in (42) to become

$$\mathbb{E}\left[Y_i(\boldsymbol{x})\right] = \mathbb{E}\left[\mathbb{E}\left[Y_i(\boldsymbol{x}) \mid \boldsymbol{X}_i = \boldsymbol{x}, U_i = u_i\right]\right]. \tag{43}$$

The aim is to calculate the average potential outcome $\mathbb{E}\left[Y(\boldsymbol{x})\right]$ for any cause $\boldsymbol{x}$. One can use Monte Carlo to estimate the right side of (43).

The deconfounder algorithm uses probabilistic factor models, including a variety of methods from Bayesian statistics and probabilistic machine learning. The causal variables are distributed according to a factor model. The probabilistic factor model of $\boldsymbol{x}$ utilizes a per-cause latent variable θ_j and a per-data point latent variable u_i, which are combined in the conditional distribution of the causes yielding

$$\begin{aligned} \theta_j &\sim P(\boldsymbol{\theta}),\ j = 1, \ldots, m, \\ u_i &\sim P(\boldsymbol{u}),\ i = 1, \ldots, n, \\ x_{ij} &\sim P(\boldsymbol{x} \mid u_i, \theta_j). \end{aligned} \tag{44}$$

Probabilistic factor models include probabilistic principal component analysis (PPCA) [135], Poisson factorization [136], non-negative matrix factorization [137]), mixed-membership models [138], [139], [140], mixture models [141] and deep generative models [142], [143], [144]. The deconfounder algorithm relies on the intuition that the factor model captures the observed dependencies among the causes $\boldsymbol{x}$ that carry information about certain confounding variables. The factor model can help correcting some of the confounding biases.

The full deconfounder algorithm consists of the following steps: Given a dataset of assigned causes and outcomes $(\boldsymbol{x}_i, Y_i)$, the goal is to compute the average potential outcome $\mathbb{E}\,[Y(\boldsymbol{x})]$ for any causes $\boldsymbol{x}$. The first step is to choose a factor model from the class in (44) and fit the model to the assigned causes $\{\boldsymbol{x}_i\}$, $i = 1, \ldots, n$. In their study, Wang and Blei [102] fitted a Poisson factorization to genomic SNPs data ($n = 5000$ and $m = 100,000$), where each SNP is represented by a count of 0, 1 or 2. The objective was to infer how the SNPs causally affect human height. This fit resulted in estimates of positive K-vectors ($k = 1, \ldots, K$) for $\boldsymbol{\theta}_j$ for each assigned cause and $\widehat{\boldsymbol{u}}_i$ for each individual i. Then, the fitted model $\widehat{M}$ is checked using posterior predictive checks [145], [146], [147], [148]. A posterior predictive check is a statistical technique used to assess the adequacy of a fitted model by comparing its predictions to the actual observed data. It is performed after fitting a Bayesian model and obtaining the posterior distribution of the model parameters. The second step is to estimate $\boldsymbol{u}_i$ as $\widehat{\boldsymbol{u}}_i = \mathbb{E}_{\widehat{M}}\,[\boldsymbol{U}_i \mid \boldsymbol{x}_i]$ for each data point if the model passes the predictive check. In Wang and Blei [102], they use the posterior predictive mean of the assigned causes as the reconstructed assignments $\widehat{x}_j(\boldsymbol{u}_i) = \widehat{\boldsymbol{u}}_i^{\mathsf{T}}\boldsymbol{\theta}_j$. The third step involves fitting an outcome model $(\tilde{M})$ to the height Y_i against both of the assigned causes $\boldsymbol{x}_i$ and reconstructed assignment $\widehat{\boldsymbol{x}}_i(\boldsymbol{u}_i)$

$$Y_i \sim \mathcal{N}(\beta_0 + \boldsymbol{\beta}^{\mathsf{T}}(\boldsymbol{x}_i - \widehat{\boldsymbol{x}}_i(\boldsymbol{u}_i)), \sigma^2), \tag{45}$$

which can be written as a linear regression

$$f(\boldsymbol{x}, \boldsymbol{x}(\boldsymbol{u})) = \beta_0 + \boldsymbol{\beta}^{\mathsf{T}}(\boldsymbol{x} - \widehat{\boldsymbol{x}}(\boldsymbol{u})). \tag{46}$$

Equation (46) is high-dimensional since $m > n$. In this step, a regularization penalty is required on $\boldsymbol{\beta}$. The next step is to perform posterior predictive checks to evaluate the fitted outcome model, as discussed in the first step. The fourth step is to estimate the average causal effect using the fitted outcome model $(\tilde{M})$ which gives an estimate of regression coefficients, $\{\beta_0, \boldsymbol{\beta}\}$, where $\boldsymbol{\beta}$ estimate the true causal effect $\boldsymbol{\beta}^*$ (in this example the coefficients of the SNPs). In each step of the deconfounder method, one can use methods such as maximum likelihood estimation (MLE), the expectation–maximization (EM) algorithm, Markov chain Monte Carlo (MCMC) and black box variational inference (BBVI) (with its RStan or Edward implementation) [102], [149], [150], [151], [152].

The main limitation of the deconfounder algorithm is that it cannot handle unobserved single-cause confounders. The algorithm also assumes that treatments cannot directly causally influence each other. However, this assumption is not generally applicable. For example, if we consider a scenario where a patient takes multiple treatments and one treatment causally influences the other treatments.

VI. CUASAL EFFECT ESTIMATION: EVALUATION METRICS AND SOFTWARE

A. METRICS FOR STANDARD CAUSAL EFFECT ESTIMATION

To assess the accuracy of causal effect estimates, various evaluation metrics are employed, for example mean absolute error (MAE), mean squared error (MSE), root mean square error (RMSE), precision in the estimation of heterogeneous effects (PEHE), and coverage probability [114], [153], [154]. Lower values of these metrics indicate better performance. In the following sections, we provide a detailed overview of these metrics, which are used to assess ATE, CATE and ATT.

1) AVERAGE TREATMENT EFFECT (ATE)

For the ATE, the following metrics are used.

Mean Absolute Error (MAE): The MAE for ATE is obtained as

$$\mathrm{MAE}_{\mathrm{ATE}} = \frac{1}{m}\sum_{j=1}^{m} |\tau_j - \hat{\tau}_j|, \tag{47}$$

where τ_j is the true ATE, $\hat{\tau}_j$ is the predicted ATE, and m represents the number of experiments.

Mean Squared Error (MSE): The MSE is calculated as

$$\mathrm{MSE}_{\mathrm{ATE}} = \frac{1}{M}\sum_{j=1}^{M} (\tau_j - \hat{\tau}_j)^2. \tag{48}$$

Root Mean Square Error (RMSE): RMSE is derived from MSE and is calculated as

$$\mathrm{RMSE}_{\mathrm{ATE}} = \sqrt{\frac{1}{M}\sum_{j=1}^{M} (\tau_j - \hat{\tau}_j)^2}. \tag{49}$$

Coverage Probability: The coverage is calculated as

$$\mathrm{Cover}_{\mathrm{ATE}} = \frac{1}{m}\sum_{j=1}^{m} \mathbf{1}(l_j < \tau_j < u_j), \tag{50}$$

where l_j and u_j are the lower and upper bounds of the interval, respectively [154].

2) CONDITIONAL AVERAGE TREATMENT EFFECT (CATE)

The following metrics are used to evaluate CATE.

Mean Absolute Error (MAE): The MAE for the CATE is

$$\mathrm{MAE}_{\mathrm{CATE}} = \frac{1}{n}\sum_{i=1}^{n}|\tau(w_i) - \hat{\tau}(w_i)|, \tag{51}$$

where $\tau(w_i)$ is the true CATE, $\hat{\tau}(w_i)$ is the predicted CATE, and n represents the number of units.

Mean Squared Error (MSE): The MSE is calculated as

$$\mathrm{MSE}_{\mathrm{CATE}} = \frac{1}{n}\sum_{i=1}^{n}(\tau(w_i) - \hat{\tau}(w_i))^2. \tag{52}$$

Root Mean Square Error (RMSE): RMSE is obtained as

$$\mathrm{RMSE}_{\mathrm{CATE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\tau(w_i) - \hat{\tau}(w_i))^2}. \tag{53}$$

Precision in Estimation of Heterogeneous Effect (PEHE): PEHE is acquired as

$$\mathrm{PEHE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i(1) - Y_i(0) - \hat{\tau}(x_i))^2, \tag{54}$$

where $Y_i(1)$ and $Y_i(0)$ are the potential outcomes.

Coverage Probability: The coverage can be calculated as

$$\mathrm{Cover}_{\mathrm{CATE}} = \frac{1}{m}\sum_{j=1}^{m}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\left(l(x_i) < \tau(x_i) < u(x_i)\right)\right). \tag{55}$$

3) AVERAGE TREATMENT EFFECT ON THE TREATED (ATT)

The following metrics are used to evaluate ATT.

Root Mean Square Error (RMSE): The RMSE for ATT is

$$\mathrm{RMSE}_{\mathrm{ATT}} = \sqrt{(\hat{\tau}_{\mathrm{att}} - \bar{\tau}_{\mathrm{att}})^2}, \tag{56}$$

where $\hat{\tau}_{\mathrm{att}}$ is the predicted ATT, and $\bar{\tau}_{\mathrm{att}}$ is the ground-truth ATT.

Coverage Probability: The coverage probability for ATT is obtained as

$$\mathrm{Cover}_{\mathrm{ATT}} = \frac{1}{m}\sum_{j=1}^{m}\mathbf{1}(l < \bar{\tau}_{\mathrm{att}} < u), \tag{57}$$

where l and u are the lower and upper bounds of the interval estimate, respectively.

B. SOFTWARE

Existing benchmarks of causal inference software packages are limited to low-dimensional data with small number of predictor variables. Hence, we do not present them here. Instead, we focus on providing an overview of software packages that can handle high-dimensional causal inference, as summarized in Table 1.

TABLE 1. Software tools for causal effect estimation.

Software Package	Language	Method
balanceHD [28]	R[a]	Approximate residual balancing for high-dimesional causal inference.
DoubleML [107]	Python and R[b]	Double machine learning for high-dimensional causal inference.
TSCI [133]	R[c]	Two stage curvature identification for causal inference with invalid instruments
bartCause [116]	R[d]	Causal inference using Bayesian additive regression trees.
CausalEGM [155]	Python and R[e]	Dimension reduction and covariate adjustment for high-dimensional causal inference.
econML [156]	Python[f]	DR, meta-learners, orthogonal random forests, IV, deep IV, and Sieve methods.
causalML [157]	Python[g]	Tree-based algorithms, meta-learner algorithms, IVs algorithms, and neural network based algorithms.
DoWhy [158]	Python[h]	DoWhy can be used for high-dimensional causal estimation by integrating with EconML and CausalML. It include: Do-calculus, back-door criterion, front-door criterion, instrumental variables (IV), mediation analysis, and propensity score based methods. Outcome model: Linear regression and generalized linear models.
causalweight [159]	R[i]	IPW, DR, and double machine learning, mediation analysis, IV, and dynamic treatment effects.
ivreg [160]	R[j]	IV: Two-stage least-squares (2SLS), two-stage-M-estimation (2SM), and 2SMM.
CausalInference [161]	R[k]	Provides a comprehensive overview of causal inference methods available in R packages.

[a] https://github.com/swager/balanceHD
[b] https://docs.doubleml.org/stable/index.html
[c] https://cran.r-project.org/web/packages/TSCI/index.html
[d] https://cran.r-project.org/web/packages/BART/index.html
[e] https://causalegm.readthedocs.io/en/latest/
[f] https://econml.azurewebsites.net/
[g] https://github.com/uber/causalml
[h] https://github.com/py-why/dowhy
[i] https://cran.r-project.org/web/packages/causalweight/index.html
[j] https://zeileis.github.io/ivreg/
[k] https://cran.r-project.org/web/views/CausalInference.html

VII. CAUSAL DISCOVERY ALGORITHMS IN HIGH DIMENSIONS

In causal inference, an established causal model is typically required to estimate causal effects. However, constructing such models beforehand is often difficult. Instead, causal discovery or learning structures from observational data has become an important method for identifying causal relationships. There are many different algorithms developed for discovering causal relationships in observational data. These algorithms output graphs containing nodes (vertices) representing random variables and directed edges representing causal relationships between them.

Causal discovery algorithms mainly depend on the following assumptions: causal sufficiency, acyclicity, causal Markov condition, and causal faithfulness condition [4], [40], [72], [162].

A. CAUSAL SUFFICIENCY

All relevant variables are observed and included in the analysis.

B. ACYCLICITY ASSUMPTION

The causal graph contains no directed cycles.

C. CAUSAL MARKOV CONDITION

Each node is conditionally independent of non-descendant nodes given its parents. This means that information about variables other than a node's parents is irrelevant for predicting its value. This simplification allows complex causal relationships to be represented as conditional independence statements within the graphical model. Hence, the causal Markov condition states that d-separation in the data-generating graph implies conditional independence in the data.

D. CAUSAL FAITHFULNESS CONDITION

Conditional independence in the data implies d-separation in the data-generating graph [163].

There are three main approaches to discovering causal relationships: constraint-based algorithms, Bayesian or score-based algorithms, and causal discovery based on functional causal models (FCMs) [164], [165], [166], [167]. To identify candidate causal graphs, the first two methods use statistical tests, while the third method uses structural equation coefficient estimation.

The Peter-Clark (PC) algorithm is a prominent constraint-based algorithm that is consistent under independent and identically distributed sampling, assuming no unobserved confounders [1]. The PC algorithm relies on the above mentioned assumptions. Given an observational dataset, PC outputs a graph called the completed partially directed acyclic graph (CPDAG), which summarizes the causal relationships among a set of variables discovered via conditional independence testing. When the conditional independence relationships are exactly equal to those implied by the DAG via d-separation, the so-called faithfulness assumption, one can learn CPDAGs from conditional independence information [168]. The PC algorithm assumes that each variable in the underlying DAG is observed and attempts to reconstruct an underlying causal graph G in three steps. The first step estimates the skeleton of G. The algorithm removes the adjacency between any two variables by starting with a fully unoriented skeleton. In the second step, the PC algorithm orients unshielded colliders. The third PC step entails applying three rules repeatedly to orient as many of the remaining undirected edges as possible. The three rules include: i) If $X - Y$, $W \rightarrow X$, and W and Y are non-adjacent, then replace $X - Y$ with $X \rightarrow Y$. ii) If $X - Y$ and $X \rightarrow W \rightarrow Y$, then replace $X - Y$ with $X \rightarrow Y$. iii) If $X - Y$, $X - W \rightarrow Y$, $X - Z \rightarrow Y$, and W and Z are non-adjacent, then replace $X - Y$ with $X \rightarrow Y$. Finally, the PC algorithm outputs a CPDAG that represents G up to its Markov equivalence class. A Markov equivalence class is a set of DAGs that represent the same set of conditional independencies. The two causal graphs G and G' belong to the same equivalence class if and only if each conditional independence that G has is also implied by G' and vice versa. In this version of the PC algorithm, as soon as conditional independence is found between two nodes, the edges are removed based on the order in which they were tested. Consequently, the output graph depends on the order of edge testing, potentially leading to different graph structures. As a solution to this limitation, a PC-stable algorithm has been proposed to remove all edges concurrently at the end of each depth test [169]. Furthermore, the PC-Max improves on this idea by selecting the p-value with the highest conditioning set, which ensures clear directionality [170]. In dense graphs, constraint-based causal algorithms have a limited runtime, which is a high-order polynomial with the maximum degree of the graph. Introducing a sparse undirected skeleton that contains all adjacencies in a data-generating DAG can improve this approach. As a solution, a mixed graphical model can be used to learn an undirected graph skeleton and avoid a large number of false positive edges [171], [172].

The PC algorithm is scalable to high-dimensional data [173], [174] and performs consistently in high-dimensional settings if the underlying DAG is sparse [175]. In such cases, the number of edges is relatively small compared to the number of variables, making the algorithm feasible and efficient.

The fast causal inference (FCI) algorithm is a generalization of the PC algorithm that can account for and sometimes discover unobserved confounding variables [1]. The FCI algorithm relaxes the causal sufficiency assumption that the data are faithful to a DAG, which may include unobservable confounders. It orients an edge similar to the PC algorithm, however, without assuming that every edge is directed in a particular direction. A partial ancestral graph (PAG) is the output of the FCI algorithm that represents the variables common to all of the DAGs in the equivalence class. For example, the end nodes of PAG can be given as: i) $X \rightarrow Y$ indicates that X is the cause of Y, ii) $X \circ \dashrightarrow Y$ indicates that X is not an ancestor of Y, iii) $X \circ - \circ Y$ indicates that no set d-separates X and Y, and iv) $X \leftrightarrow Y$ indicates that there is at least one unobserved common cause of X and Y.

The FCI is also scalable to high-dimensional data [176]. However, it is only consistent in a high-dimensional setting if the so-called Possible-D-SEP sets are sparse [175], [177]. For computational speed improvement, [168] developed the really fast causal inference (RFCI) algorithm, which is significantly faster than FCI. RFCI differs from FCI by not performing conditional independence tests on subsets of possible d-separate sets, instead performing additional tests before orienting structures. RFCI scales similarly to FCI in high-dimensional settings, but is faster.

Since a constraint-based search algorithm performs multiple tests, controlling for false positives is crucial. The standard approach to dealing with multiple hypothesis testing problems is to control the proportion of false positives among the rejected null hypotheses using the false discovery rate (FDR) method [178], [179]. An FDR control procedure takes as input a desired FDR level q and a set of p-values. Here, the q-values are the adjusted p-values found using an optimized FDR approach. The procedure then returns a significance level α for the set of p-values. Strobl et al. [178] proposed the PC with p-values (PC-p), an algorithm that computes edge-specific p-values. The PC-p algorithm uses the p-values returned by many conditional independence (CI) tests to upper bound the p-values of more complex edge-specific hypothesis tests. It controls the false discovery rate (FDR) using these bounded p-values.

The FDR at threshold α is the expected proportion of false positives among the rejected null hypotheses, defined as

$$\mathrm{FDR}(\alpha) \triangleq \mathbb{E}\left[\frac{V}{\max\{R, 1\}}\right], \tag{58}$$

where V is the number of false positives, R is the total number of null hypotheses rejected, and $\max\{R, 1\}$ is the realized *FDR* at α. Benjamini and Yekutieli [180] proposed the following FDR estimator for m hypothesis tests,

$$\widehat{\mathrm{FDR}}(\alpha) \triangleq \frac{m\alpha \sum_{i=1}^{m} \frac{1}{i}}{\max\{R, 1\}}. \tag{59}$$

The optimal threshold α^* that achieves strong control of the FDR is obtained as

$$\alpha^* = \arg\max_{\alpha}\{\widehat{\mathrm{FDR}}(\alpha) \leq q\}. \tag{60}$$

The FDR controlling procedure based on $\widehat{\mathrm{FDR}}$ involves the rejection of all null hypotheses with p-values below the α^* threshold.

For the de-biased LASSO Granger causal estimator, Chaudhry et al. [181] proposed FDR and false discovery proportion (FDP) following

$$\mathrm{FDP}(\nu) = \frac{\sum_{i\in H_0} \mathbf{1}|t_i| \geq \nu}{\max\{\sum_{1\leq j\leq lk} \mathbf{1}|t_j| \geq \nu, 1\}}, \tag{61}$$

where H_0 is the null hypothesis, t is the test statistic, l is the maximum lag and k is the number of parameters. One can choose ν as

$$\widehat{\nu} = \inf\left\{\nu > 0 : \frac{\sum_{i\in H_0} \mathbf{1}|t_i| \geq \nu}{\max\{\sum_{1\leq j\leq lk} \mathbf{1}|t_j| \geq \nu, 1\}} \leq \alpha\right\} \tag{62}$$

such that

$$\mathrm{FDR}(\widehat{\nu}) = \mathbb{E}\left[\mathrm{FDP}(\widehat{\nu})\right] \leq \alpha. \tag{63}$$

The null distribution is a crucial tool in hypothesis testing, especially in controlling the rate of false positives. However, in high-dimensional settings, the theoretical null distribution for FDR may not accurately reflect the distribution of test statistics under the null hypothesis. This may lead to incorrect scientific conclusions, such as false positives or false negatives. To overcome this issue, Efron [18] proposes the use of empirical estimation methods, such as permutation tests or bootstrapping to estimate the null distribution.

In contrast to the PC and FCI algorithms, the greedy equivalence search (GES) is a score-based algorithm [48]. Instead of conditional independence tests, this method infers causal structure by maximizing a likelihood score. To produce an output DAG, GES begins with an empty graph, then adds or removes edges one at a time. It will keep changing patterns until it can no longer make any more changes. Based on the relative strengths of associations and conditional dependencies among the variables, the GES procedure estimates the connections between them. A decision is made throughout the algorithm whether adding a directed edge to a graph will improve a fit based on a locally consistent score such as the Bayesian information criterion (BIC) [48]. The BIC score approximates the posterior probability of the model given the data. The smaller value indicates a better fit for the model. Then the edge that gives the best fit is added. An improved version of GES, known as the fast greedy equivalence Search (FGES) algorithm, uses the same scoring system and search algorithm but is much faster than GES [182], [183]. It is also scalable to high-dimensional data [176].

Several approaches built on functional causal models (FCMs) have been proposed for causal discovery using the causal sufficiency assumption from continuous variables. In an FCM, the effect on Y is modeled as a function of the direct causes X and some unobserved noise as $Y = f(X, U_Y)$, where the noise term U_Y is independent of X [4]. As an example, the linear non-Gaussian model (LiNGaM) algorithms can recover DAG structures from data generated with non-Gaussian noise using independent components analysis (ICA) to uniquely identify the causal structure [164], [165].

In contrast to constraint-based and score-based algorithms, these algorithms do not rely on the faithfulness assumption [165]. Furthermore, Zeng et al. [184] scaled FCMs to high-dimensional deterministic model (HDDM).

VIII. CUASAL DISCOVERY: EVALUATION METRICS AND SOFTWARE

A. EVALUATION METRICS

There are two primary types of metrics used to evaluate causal discovery algorithms: 1) the distance between the inferred causal graph and the ground truth, and 2) the accuracy of the identified causal relationships.

1) STRUCTURAL HAMMING DISTANCE (SHD)

SHD measures the difference between two causal graphs: the ground-truth graph and a predicted graph. It is defined as the number of edits required to make the predicted graph match the ground-truth graph. These edits can include adding, deleting, or reorienting edges [185]. The formula for SHD is:

$$\mathrm{SHD} = E_A + E_D + E_I, \tag{64}$$

where E_A is the number of edges added, E_D is the number of edges deleted, and E_I is the number of incorrectly oriented edges.

2) FROBENIUS NORM

The Frobenius norm measures the distance between the true adjacency matrix $\mathcal{B}$ and its prediction $\widehat{\mathcal{B}}$ [186]. This can be expressed as

$$\sqrt{\text{trace}\left((\mathcal{B}-\widehat{\mathcal{B}})^T(\mathcal{B}-\widehat{\mathcal{B}})\right)}. \tag{65}$$

3) STRUCTURAL INTERVENTION DISTANCE (SID)

Given a true DAG G and an estimate $\hat{G}$, SID counts the number of intervention distributions, which are computed using the structure of $\widehat{G}$, that coincide with the true intervention distributions inferred from G. More specifically, SID measures the pairs of vertices (i, j) for which G accurately predicts intervention distributions within its Markov equivalence class. This can be expressed as: Let $\mathcal{G}$ be the space of DAGs with p variables. Then, the SID can be defind as

$$\begin{aligned} &\text{SID} : \mathcal{G} \times \mathcal{G} \to \mathbb{N}, \\ &(G, \widehat{G}) \mapsto G^*, \end{aligned} \tag{66}$$

where $G^* = \#\{(i, j), i \neq j \mid$ the intervention distribution from i to j is falsely estimated by $\widehat{G}$ with respect to $G\}$ [187].

4) PRECISION, RECALL, AND ACCURACY

These metrics evaluate the correspondence between predicted and true adjacencies, as well as between predicted and true arrows [163], [183]. Here, two nodes are considered adjacent if there is any type of edge (directed or undirected) between them, whereas two nodes are connected by arrows if there is a directed edge between them in either direction.

a: PRECISION

Also known as positive predictive value, measures the accuracy of the algorithm's positive predictions. It is defined as the ratio of true positives (TP) to the sum of true positives (TP) and false positives (FP): precision $= \frac{TP}{TP+FP}$.

b: RECALL

Also known as sensitivity or true positive rate (TPR), it measures the proportion of actual positive instances that the algorithm correctly identifies. It is defined as the ratio of true positives (TP) to the sum of true positives (TP) and false negatives (FN): recall $= \frac{TP}{TP+FN}$. The optimal algorithm will have a precision and recall close to 100% [163], [183], [188]. Hence, precision and recall assess how effectively the model distinguishes between the two classes. In addition, accuracy is another important metric defined as accuracy $= \frac{TP+TN}{TP+FP+FN+TN}$. In this context, the false discovery rate (FDR) is also a relevant metric, which can be expressed as FDR $= \frac{FP}{FP+TP}$.

c: TRADE-OFF

Precision and recall often have an inverse relationship, meaning that improving one typically reduces the other. For example, a model with high precision is very selective about what it classifies as positive, resulting in fewer false positives but potentially missing some true positives. Conversely, a model with high recall identifies more actual positive cases, but this increase in sensitivity can also lead to a higher number of false positives. To balance precision and recall, metrics like the F1 score can be used [189].

d: F1 SCORE

The F1 score is the harmonic mean of precision and recall, given by: F1 Score $= 2\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} = \frac{2TP}{FN+FP+2TP}$. The F1 score is useful when one need a single metric to evaluate the performance of a model, especially when one want to balance both precision and recall. **Area Under the Curve (AUC):** AUC measures the area under the Receiver Operating Characteristic (ROC) curve, which plots the false positive rate (FPR) against the TPR (recall). Here, FPR is defined as FPR $= \frac{FP}{FP+TN}$. A higher AUC value indicates that the model is better at distinguishing between the classes. An optimal model achieves an AUC close to 1.

TABLE 2. Software tools for causal discovery.

Software Package	Language	Method
pcalg [175]	R[a]	PC, FCI, RFCI, GIES, IDA, generalized backdoor criterion (GBC), and generalized adjustment criterion (GAC).
bnlearn [190]	R[b]	Constraint-based algorithms, score-based algorithms, hybrid, local discovery algorithms,and Bayesian network classifiers.
causal-learn [191]	Python[c]	Constraint-based algorithms: PC, FCI, CD-NOD. Score-based algorithms: GES with the BIC score or generalized score, and exact search. Functional causal models (FCMs): LiNGAM-based methods, post-nonlinear causal models, and additive noise models. Hidden causal representation learning: Generalized independence noise (GIN). Permutation-based causal discovery methods: GRaSP, and Granger causality.
Tetrad [192]	Python and R[d]	PC, PC stable, FCI, GFCI and FGES.
CausalDiscovery [193]	Python[e]	Causal discovery algorithms including PC, GES, GIES, and LiNGAM. Metrics: Precision, recall, SHD, and SID.
CausalInference.jl	Julia[f]	PC, GES, FCI, exact score-based method and covariate adjustment sets for estimating causal effects.

[a] https://cran.r-project.org/web/packages/pcalg/index.html
[b] https://www.bnlearn.com/
[c] https://causal-learn.readthedocs.io/en/latest/
[d] https://www.cmu.edu/dietrich/philosophy/tetrad/use-tetrad/
[e] https://github.com/FenTechSolutions/CausalDiscoveryToolbox
[f] https://mschauer.github.io/CausalInference.jl/latest/

B. SOFTWARE

As with causal inference, existing benchmarks of software packages for causal discovery are also limited to low-dimensional data. We provide an overview of software packages that can handle high-dimensional causal discovery in Table 2.

IX. CONCLUSION AND OPEN PROBLEMS

Confounding is the main issue that can be misleading when one aims at estimating causal effects and discovering causal structures from observational studies. Confounding bias can be a spurious correlation in high-dimensional data that leads to wrong causal discoveries and incorrect causal effect estimation. An investigator can to some extent control spurious correlations by adjusting the proportion of false positives. This can be done using a false discovery rate (FDR) in both estimating causal effects and discovering causal relationships by using reasonable thresholds. High-dimensional complications such as endogeneity, noise accumulation and computational difficulties associated with covariance estimation affect both the estimation of causal effects and the discovery of causal relationships. In contrast to spurious correlations, incidental endogeneity refers to a genuine correlation between a variable and the residuals. As a solution, one can use the instrumental variables via two-stage least squares or over-identification methods with penalized regression model.

Noise accumulation poses a difficulty for causal discovery algorithms as it occurs when multiple parameters are estimated or tested at the same time, resulting in estimation errors. It attenuates true dependencies and lowers detection power. To overcome the limitations of noise accumulation and improve performance, it is advisable to utilize an algorithm that employs a strategy of conducting multiple tests while using conditioning sets of lower dimensions.

An estimation of the covariance matrix that encodes the conditional independence of variables can also be used to discover causal relationships from a DAG. In high-dimensional graphical models, the graphs tend to be dense, making the computation of the covariance matrix not only demanding but also resulting in the sample covariance matrix being a poor approximation of the empirical covariance matrix. Therefore, a sparse estimation of the covariance matrix with a small number of edges between nodes is necessary. This can be achieved by using penalized likelihood with an L_1-norm, e.g., LASSO penalty on the entries of the covariance matrix. Sundaramoorthy et al. [194] proposed an L_0-norm constraint within the framework of the greedy sparse simplex (GSS) algorithm, as the L_0-norm promotes better sparsity compared to the L_1-norm. However, the L_0-norm induces a non-convex optimization problem that is NP-hard, limiting its applicability to relatively small sample sizes.

Checking the validity of instrumental variables in high-dimensional data is almost impossible. For example, in Mendelian randomization studies, genetic variants are used as instruments. Complete knowledge of instrument validity is equivalent to complete knowledge of the functions of the involved genes. Hence, we can only appreciate the proposed method for estimating causal effects in the absence of complete knowledge of the validity of all instrumental variables [131], [133].

Recently, Liu et al. [155] introduced an approach called causal encoding generative modeling (CausalEGM) to handle high-dimensional covariates through a dependency-aware dimension reduction strategy. The main focus of this method is to identify unobserved confounders that affect both the treatment and the outcome and that are used for covariate adjustment. However, the main challenge under the CausalEGM framework lies in handling this through instrumental variables, as the validity of these instruments remains an open problem.

Since the quality of observational studies is often assessed in terms of how closely it approximates a randomized controlled trial, it is natural to wonder how to combine experimental and observational data for causal inference. The subjects in the experimental domain are only followed for a short period of time. Hence, the long-term effects of treatment are unobserved but short-term effects will be observed. In the observational domain, both short-term effects and long-term effects of treatments are observed but subject to unobserved confounders. Athey et al. [7] and Ghassami et al. [8] proposed a method for systematically combining such data to estimate treatment effects.

Pearl [195] stressed the importance of including causal reasoning in machine learning to enhance their accuracy in predicting outcomes and to make them easier to interpret. We emphasize the combination of causal reasoning and machine learning tools as essential for causal effect estimation and causal discovery. These approaches allow new models, methods, and algorithms to be implemented efficiently into the existing causal framework and extended to high-dimensional settings. Finally, we present some specific open problems and possible future developments.

A. CAUSAL INFERENCE IN HIGH-DIMENSIONAL GENOMICS

In recent years, the field of precision medicine has emerged as a transformative approach to healthcare. Precision medicine tailors treatments to individual patients based on their unique genetic makeup, lifestyle, and environment. By understanding the specific factors that contribute to a person's health, doctors can provide more effective and personalized care. High-throughput genomic technologies generate vast amounts of data on genetic variations, including single nucleotide polymorphism (SNP) markers. However, data is often obtained from a limited number of patient samples. The goal is to identify which of these genomic variants that directly cause specific diseases, and then, based on the findings, develop personalized medicine and care. However, traditional causal methods are often inadequate for high-dimensional genomic datasets, where the number of genomic variables exceeds the sample size.

Recent methods address the challenge of high-dimensional data through dimensionality reduction and regularization. Dimensionality reduction transforms data into a lower-dimensional space, while regularization adds a penalty to reduce the number of variables, thereby enhancing model generalization, variable selection, and interpretability. However, both methods have drawbacks: dimensionality reduction leads to the loss of the original variables and their effects, complicating the understanding of variable relationships, which is critical in fields like genetics. Regularization, on the other hand, often shrinks causal effect estimates, potentially leading to biased results and conclusions [28]. Hence, developing novel causal inference methods for high-dimensional genomics is crucial. Currently, there is considerable interest in nonconvex and nonsmooth models that effectively reduce estimation bias [196]. We recommend that the computational challenges associated with nonconvex and nonsmooth problems be tackled using advanced optimization techniques such as the alternating direction method of multipliers (ADMM) and the Peaceman-Rachford splitting method (PRSM) with proximal operators [197].

B. CAUSAL DISCOVERY METHODS FOR NON-TABULAR DATA

Developing causal discovery methods for non-tabular (also referred to as unstructured) data represents a critical frontier in the field of causal learning [198], [199]. The capacity to extract causal insights from non-tabular data sources such as text, images, and audio will unlock new opportunities across various disciplines, offering a more comprehensive and accurate understanding of cause-effect relationships in real-world scenarios. For example, in healthcare, these methods can be used to analyse medical images, patient records, or clinical notes to uncover the underlying causes of diseases or assess the effectiveness of treatments. In social media analysis, causal discovery helps researchers understand how specific types of content influence public opinion, consumer behavior, or even political trends.

For example, Lopez-Paz et al. [200] propose a new algorithm, the neural causation coefficient (NCC), capable of discovering causal signals in images. Similarly, Li et al. [201] introduce the visual causal discovery network (V-CDN), which can discover causal relationships from videos in an end-to-end fashion without supervision on the ground-truth graph structure.

ACKNOWLEDGMENT

The authors thank the three reviewers for their thoughtful feedback and constructive comments.

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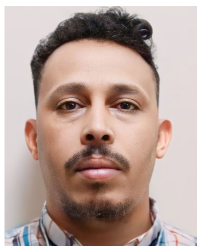

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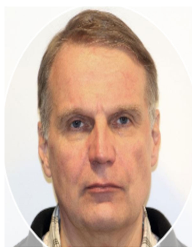

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